

POLITICIZED SCIENTISTS:
CREDIBILITY COST OF POLITICAL EXPRESSION ON TWITTER

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As social media becomes prominent within academia, we examine its reputational costs for academics. Analyzing Twitter posts from 98,000 scientists (2016–22), we uncover substantial political expression. Online experiments with 6,000 U.S. respondents and 135 journalists, rating synthetic academic profiles with different political affiliations, reveal that politically neutral scientists are seen as the most credible. Strikingly, political expressions result in monotonic penalties: Stronger posts reduce perceived credibility of scientists and their research and audience engagement more, particularly among oppositely aligned respondents. Two surveys with scientists highlight their awareness to penalties, their perceived benefits, and a consensus on limiting political expression outside their expertise.

KEYWORDS: Twitter, Scientists' Credibility, Polarization, Online Experiment.

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IRB approval was obtained from WZB Berlin (No. 235), and the studies were pre-registered on AsPredicted (Nos. [166935](#), [179009](#), [181452](#), [186295](#), [206927](#), and [235820](#)). Financial support was provided by the University of Bath, WZB, and Imperial College London; funders had no role in study design, data collection and analysis, publication decisions, or manuscript preparation. We thank Nicolas Ajzenman, Aurelien Baillon, Kai Baron, David Beavers, Sascha Becker, George Beknazar-Yuzbashev, Apurav Yash Bhatiya, Luca Braghieri, Abel Brodeur, Valeria Burdea, Nathan Canen, Jacob Conway, Guillermo Cruces, Lachlan Deer, Mirko Draca, Anna Dreber, Ruben Durante, Thiemo Fetzner, Andy Guess, Felix Holzmeister, Steffen Huck, Rafael Jimenez-Duran, Alexey Makarin, Giuseppe Musillo, Andrew Oswald, Tom Palfrey, Barbara Piotrowska, Martina Pocchiari, Pedro Rey-Biel, Jaideep Roy, Chris Roth, Carlo Schwarz, Mateusz Stalinski, Andreas Stegmann, Jean Tirole, Egon Tripodi and seminar and conference participants at Innsbruck, NHH, UAB, PEPES, EWMES24, CODE, AFE24, MWZ, Liverpool, Lancaster, EAYE, Women in PolEcon, Petralia, Bari Applied Workshop, ZBW Open Science, Narratives Memory and Beliefs Workshop, CEPR St'Gallen. The authors declare no competing interests.

1. INTRODUCTION

In a "post-truth" era, trust in science is a cornerstone of informed decision-making and effective public policy (McIntyre, 2018, Angelucci and Prat, 2024, Bursztyn et al., 2023b, Arold, 2024, Ash et al., 2024). The COVID-19 pandemic highlighted this reality, illustrating how confidence in scientific expertise shapes public health responses. Similarly, climate skepticism hinders progress toward environmental goals.¹ Skepticism extends beyond these domains, affecting economic development, education, and societal trust more broadly.

Credibility is essential to science, yet public trust is at a relative low, particularly among U.S. conservatives.² A key driver of distrust is the perception of political bias among scientists (Altenmüller et al., 2024), amplified by the accessibility of their social media activity, where nearly half of academic content engagement comes from non-academics (Mohammedi et al., 2018, Maleki and Holmberg, 2023).

This paper proceeds in two parts. First, we use Large Language Models (LLMs) to measure the extent and nuance of scientists' political expression on social media. Second, we assess whether their online political engagement affects public perceptions, focusing on the reputational costs for scientists as public opinion makers (Aina, 2023) shaping individual preferences and behaviors (Algan et al., 2021, Burnitt et al., 2024, Martinez-Bravo and Stegmann, 2022). Using several online experiments, we evaluate how revealing scientists' political views on social media influences perceptions of their credibility and contributes to audience polarization.

¹Higher trust in science is linked to earlier adoption of preventive measures during COVID-19 (Algan et al., 2021, Allcott et al., 2020, Bartoš et al., 2022, Bowles et al., 2023, Eichengreen et al., 2021), whereas skepticism has driven reliance on unverified remedies, such as those documented in Argentina (Calónico et al., 2023, Albornoz et al., 2024). On climate change, Druckman and McGrath (2019) highlights how differing beliefs about credible evidence influence policy support.

²Nichols (2017) discusses hostility toward expertise in the U.S., while Lupia et al. (2024) documents declining trust in science among U.S. respondents. Factors include misinformation (West and Bergstrom, 2021, Roozenbeek et al., 2020), historical failings (Scharff et al., 2010), the reproducibility crisis (Hendriks et al., 2020), conspiracy theories (Rutjens and Večkalov, 2022, Douglas, 2021), science-related populism (Mede and Schäfer, 2020, Mede et al., 2021), and political ideology (Cologna et al., 2025), with U.S. conservatives reporting lower trust in scientists, stronger anti-science attitudes, and less confidence in their intentions and methods (Mede, 2022, Funk et al., 2020, Li and Qian, 2022, Azevedo and Jost, 2021).

Our findings reveal evidence of both *ideological* polarization (divergent political positions among academics) and *affective* polarization (public aversion to scientists associated with opposing partisan groups) (Lelkes, 2016, Barberá, 2020). The central questions of our study are: *To what extent do scientists express polarized political opinions on social media? And what impact does this political expression have on their perceived credibility?*

Impact is increasingly crucial for academics, with social media playing a growing role in research dissemination and public engagement with science. On the one hand, studies show that social media engagement can benefit academics (Chan et al., 2023, Klar et al., 2020, Qiu et al., 2024, Boken et al., 2023), and Altmetric data reveal a sharp rise in the online presence of scientific research on social media between 2011 and 2020.³ On the other hand, recent cross-country evidence shows that the public primarily encounters science on social media, frequently discusses it, and often takes outspoken positions—sometimes even participating in rallies related to scientific issues, particularly in the United States (Mede et al., 2025). These trends highlight the increasing interconnection between science and politics, as well as the visibility of scientific discourse in the public sphere. Our study extends this perspective by examining how scientists use social media not just to share research, but to engage in politically salient debates—with implications for credibility and public polarization.

First, we build on Garg and Fetzer (2025b) by analyzing the extent of online political discourse and issue-based disagreement among U.S. scientists (university-based researchers) compared to general social media users. Examining political slants on social media offers three key advantages: (i) capturing a large, cross-disciplinary sample of academics, (ii) providing a nuanced view of ideological positions beyond a left-right spectrum, and (iii) focusing on publicly visible content, avoiding the limitations of survey-based or publication-based classifications.

Our descriptive analysis reveals that 44% of 97,737 U.S. academics on Twitter expressed political opinions by making at least one non-neutral post on any of five politically salient topics—(i) abortion rights, (ii) climate action, (iii) immigration, (iv) income redistribution,

³While this trend partly reflects overall growth in social media use, our data suggest a specific surge in the visibility of scientific research. Altmetric tracks research impact across newspapers, blogs, social media, policy briefs, patents, and academic citations.

and (v) racial equality—between 2016 and 2022, compared to only 7% of the general non-academic U.S. Twitter base.^{4,5} Political engagement is widespread across disciplines and topics. Notably, although 29% of academics’ topical tweets reference research, only 7% of research tweets mention these topics, suggesting a selective overlap. Academics tend to be more neutral when tweeting about research, and more explicit when engaging with politically salient issues. Consistent with this, while many non-neutral tweets are grounded in verifiable facts, 51% of topical tweets contain no checkable claim. Political discourse among academics is thus predominantly *opinion-based* rather than purely informational.

Original posts are also less explicit than retweets, and high-reach academics tend to adopt more moderate stances.⁶ Positions have also shifted over time, with racial equality standing out: disagreement among academics is both more pronounced and widening compared to general users.⁷

According to Morris (2001), engaging in “politically incorrect” communication can result in reputational loss, as audiences may draw adverse inferences about the speaker. Since such reputational concerns can lead to information loss, the second part of our paper quantifies this cost through an online experiment with a representative sample of 1,700 U.S. respondents.⁸ They assessed each scientist’s personal credibility, the credibility of their re-

⁴These topics are identified as the most salient in U.S. political debates by Gallup and Pew Research.

⁵A natural benchmark is Garg and Fetzer (2025b), who study the same five issues in a balanced global panel of 99,274 scholars (tweeting in both 2016:H1 and 2022:H2). In that sample, 27% of scholars ever expressed a non-neutral view. The higher U.S. share (44%) places American academics at the top of the international distribution. Sampling differs: their design excludes accounts that joined after 2016 or left before 2022 to enable within-author comparisons; our goal is to characterize the full stock of U.S. academics visible to survey respondents (Section 4).

⁶These patterns highlight the sensitivity of our classification in capturing the intensity of political expression.

⁷This trend of increased political activism among scientists has been linked to events like the March for Science (Russell and Tegelberg, 2020, Campbell et al., 2023).

⁸The broader welfare implications of expert communication are complex. Chakraborty et al. (2020) distinguish between partisan endorsements (post-platform choice) and policy advocacy (pre-platform choice). While endorsements may reduce voter welfare by distorting platforms, these distortions can incentivize experts to provide informative policy advice—unless the conflict between expert and voter preferences is too large. Moreover, despite potential reputational costs, scientists can also benefit professionally from social media, including citation premiums (Chan et al., 2023, Klar et al., 2020) and improved recruitment outcomes in the academic job market (Qiu et al., 2024).

search, and their willingness to engage with their content. This conjoint design allows us to isolate and precisely estimate the effects of political expression on public perceptions, offering more experimental control than a Twitter-based approach.⁹

Findings reveal a significant *credibility penalty* for scientists who engage in political discourse, regardless of political orientation. Scientists expressing either left- or right-leaning views face a monotonic decline in *personal credibility* (first outcome): Strong Republicans are rated 39% less credible than neutral scientists, Strong Democrats 11% less; Moderate Republicans and Democrats are rated 9% and 7% less credible, respectively. This penalty extends to the *credibility of their academic work* (second outcome), indicating that reputational costs go beyond personal perception and exceed standard in-group bias. It also translates into lower public *willingness to engage with their work* (third outcome): respondents are 42% less willing to read opinion pieces from Strong Republicans, 10.7% less for Strong Democrats, and 8% and 4.5% less for Moderate Republicans and Democrats, respectively, compared to neutral scientists. All effects are statistically significant at the 1% level.

Credibility penalties reflect a strong partisan bias: respondents mainly perceive scientists aligned with opposing political views as less credible. Democrat respondents rate neutral and Democrat-leaning scientists similarly but lose trust in Republican-leaning ones. Republican respondents generally prefer moderately Republican scientists to neutral, while still penalizing strongly conservative profiles, though less so than they penalize Democrat-leaning scientists. These patterns suggest that scientists' political discourse can serve as a channel for polarizing public perceptions and reducing engagement with scientific content, depending on audience ideology. Both set of results can be rationalized with models of conformity (Bernheim, 1994, Orzach, 2024).

Among other tested attributes, scientists from highly ranked institutions and those in senior positions are perceived as more credible, while gender and field have no significant effect. Notably, political affiliation emerges as the most salient factor shaping public perceptions of scientists.

⁹Conjoint designs have been used to study medical decisions (Chan, 2022), repugnant transactions (Elías et al., 2019, Sullivan, 2021), financial choices (Macchi, 2023), dating preferences (Low, 2014), and charitable donations (List and Lucking-Reiley, 2002).

We conducted several robustness checks. First, we replicated our main findings with a separate sample of 2,000 U.S. respondents, confirming that results hold under a different outcome structure, as well as with an additional sample of 2,000 U.S. respondents who report being Twitter users. We also found minimal carryover effects across profiles, and results remain stable after excluding “speeders” and adjusting for multiple hypothesis testing. A permutation test and a validation task with new respondents confirmed that participants accurately identified the intended political affiliations of scientists in the vignettes. Finally, we conducted an additional robustness experiment to bound potential experimenter demand effects, in line with the approach of [de Quidt et al. \(2018\)](#), and found that any such influence is minimal and cannot account for our results.

To assess external validity, we tested whether the *credibility penalty* for politically engaged scientists extends to journalists—an important audience, as they regularly cover scientific research and interview scientists ([Alabrese, 2022](#)). Among 135 international journalists surveyed, we observed clear asymmetries: Republican scientists were rated 33% less credible than neutral ones, and their opinion pieces were 40% less likely to be included in newsletters. By contrast, Democrat scientists were rated only 5% less credible and 2% less likely to be featured—effects that are statistically indistinguishable from zero, likely due to the overrepresentation of liberal journalists. Partisan bias was evident: Liberal journalists were 7, 9, and 5 times less likely to rate a Republican scientist’s credibility, research, and newsletter inclusion favorably compared to a Democrat. Conservative journalists showed a comparable bias in the opposite direction, being 4, 9, and 7 times less likely to favor Democrat scientists across the same dimensions.¹⁰

According to the sender-receiver framework in [Gentzkow and Shapiro \(2006\)](#), a sender who fails to align their message with the receiver’s prior beliefs incurs a greater credibility penalty—especially when the quality of the sender cannot be easily verified ex post. For scientists, this penalty may arise in two ways: when discussing politically charged research topics that conflict with readers’ views (leading audiences to infer political bias) or when signaling a political identity cue unrelated to research (which may elicit affective polarization). To investigate these channels, we conducted an additional experimental task after the

¹⁰This aligns with findings from [Boxell and Conway \(2022\)](#), who show that 16% of slant variation across outlets can be explained by journalist preferences.

conjoint experiment. This between-subject task isolates the credibility effects of (i) tweeting about politically salient research that may implicitly signal ideological bias versus (ii) explicitly signaling political affiliation through unrelated political cues. We held the research message constant, varied the political signal in an otherwise uninformative bio, and restricted all profiles to economists to keep expertise constant.

Both sharing politically salient research and signaling political identity independently shape public perceptions. First, we find that merely tweeting about politically related research influences credibility: Democrats view economists more favorably when the research aligns with their political views, while Republicans rate economists lower when the research content is misaligned. These credibility shifts occur even in the absence of any explicit political signaling. Second, adding a political identity cue amplifies these effects. Among Democrats, a left-leaning bio further increases credibility, while a right-leaning one sharply reduces it. For Republicans, the pattern reverses: left-leaning signals exacerbate the credibility penalty for misaligned research, while right-leaning signals moderate it. These effects extend to newsletter demand—a behavioral measure of audience engagement curated following (Chopra et al., 2022, 2024)—where alignment between the audience’s political views and the scientist’s signal strongly predicts sign-ups. Together, these findings confirm that both inferred ideological bias (based on research content) and affective polarization (based on identity cues) contribute to audience reactions, shaping who the public trusts and engages with.

What do scientists think? We surveyed 128 international scientists on Prolific to assess whether they anticipate the credibility penalty observed in our experiment and to understand their views on political expression online. First, scientists expected a larger penalty than what we observed and considered it more acceptable to express political opinions within their area of expertise than outside it: a norm they believe to be widely shared among other academics. Second, many reported hesitating to share political views on social media, suggesting a potential for information loss in public discourse (Morris, 2001, Ottaviani and Sørensen, 2006). Third, while they anticipated mild reputational costs for commenting on topics outside their field, they expected modest reputational benefits when engaging within their expertise. However, perceptions of costs and benefits largely overlapped. To examine this further, we conducted a follow-up survey with 118 additional international scientists. Respondents perceived clear professional benefits of political expression for media appear-

ances and policy roles, smaller benefits for network size and citations, and ambiguous or slight costs for top journal publications and academic job offers.

To probe a mechanism behind the credibility penalty, consistent with the perceived benefits scientists report, we field an additional experiment that makes a scientist's *visibility incentive* explicit. Adapting the design of [Bursztyn et al. \(2023a\)](#), half of the respondents, before viewing a scientist's tweet, also receive the message that *we (the researchers)* will promote the post and that the scientist knew this *ex ante*. This manipulation reveals no uniform "opportunism tax." Instead, updating is partisan. Among Democrats, the promotion information amplifies our main pattern: higher credibility for authors of aligned (left-leaning) posts and lower credibility for authors of misaligned (right-leaning) posts. Among Republicans, promotion raises credibility for authors of aligned (right-leaning) posts and attenuates the penalty for authors of misaligned (left-leaning) posts; neutral promotion is disliked. General trust in science shifts in the same direction under promotion (positive when aligned for Democrats; positive for both left- and right-leaning content among Republicans). Taken together, these results indicate that audiences recognize scientists' incentives but interpret them through partisan congruence and normative expectations about scientists' public engagement.

Overall, studying the reputational costs of scientists' online political engagement reveals a fundamental trade-off. Expressing political views can undermine scientists' credibility, limiting their ability to shape public preferences and influence decision-making while potentially deepening affective polarization. At the same time, anticipating this credibility penalty may discourage scientists from participating in policy debates, leading to information loss and diminishing the societal value of expert engagement. These effects are further mediated by scientists' perceptions of personal career benefits from engaging on social media.

Contribution to the Literature Our findings deepen our understanding of how the public perceives scientists and their communication ([Altenmüller et al., 2024](#), [Blastland et al., 2020](#), [Norris, 2020](#)). Prior work has explored how uncertainty and political transparency affect trust in science ([Van Der Bles et al., 2019, 2020](#), [Petersen et al., 2021](#)). For example, [Kotcher et al. \(2017\)](#) found that climate scientists' advocacy generally did not reduce credibility, except when promoting nuclear power. [Zhang \(2023\)](#) showed that political en-

dorsements by scientific journals, such as Nature’s support for Joe Biden, can polarize perceptions of publishers and science more broadly, particularly among Republicans.

We extend [Zhang \(2023\)](#) in key ways. We provide the first evidence of a visibility–credibility trade-off in scientists’ political expression, combining (i) observational data on scientists’ political engagement on Twitter, (ii) surveys of international scientists, and (iii) online experiments with representative samples of U.S. respondents, Twitter users, and journalists. Rather than focusing on the editorial decision of a single publisher, which may reflect institutional rather than individual scientists’ preferences, we provide large-scale observational evidence on scientists’ own political discourse across disciplines and over time, and causally identify the reputational effects of scientists’ political expression through experiments.

More broadly, we contribute to the literature on the communication (or absence) of politically salient topics in the U.S., including the “spiral of silence” ([Huang and Ho, 2023](#)), both offline ([Braghieri, 2024](#)) and online ([Bursztyn et al., 2023a](#)). This literature focuses on decisions to voice controversial opinions and their downstream effects. To our knowledge, this is the first study to examine U.S. academics’ online communication on politically charged topics and its impact on public perceptions. We also contribute to work on politically motivated reasoning in the acquisition ([Chopra et al., 2024](#), [Faia et al., 2024](#)) and interpretation ([Gentzkow et al., 2023](#), [Thaler, 2024](#)) of information, by showing how political affiliation mediates responses to scientists’ views.

We further relate to the growing literature using text-as-data to measure political ideology from sources such as official documents ([Ash, 2016](#), [Hansen et al., 2018](#), [Grimmer, 2010](#)), online news ([Cagé et al., 2020](#)), product descriptions ([Fetzer et al., 2024](#)), political speeches ([Jensen et al., 2012](#), [Gentzkow et al., 2019](#)), academic publications ([Garg and Fetzer, 2025a](#), [Jelveh et al., 2024](#), [Brown and Gupta, 2023](#)), and surveys ([Draca and Schwarz, 2024](#)). Our study builds directly on [Garg and Fetzer \(2025b\)](#), who show that scientists’ political expression differs from that of the general public in both topic and tone, with public narratives often shaped by highly visible but less academically impactful scientists. We extend this work by analyzing polarization in political discourse among U.S. academics and causally identifying the reputational risks of political engagement, which vary sharply by audience ideology.

Our findings contribute to broader evidence on ideological and affective polarization in the U.S., documenting how echo chambers can deepen public divides (Canen et al., 2020, 2021, Fiorina and Abrams, 2008, Alesina et al., 2020, Iyengar et al., 2019, Gentzkow and Shapiro, 2011, Levy, 2021, Chopra et al., 2024, Mosleh et al., 2021, Colleoni et al., 2014, Flaxman et al., 2016, Stewart et al., 2019, Boxell et al., 2024, Kahan et al., 2011). We find that individuals gravitate toward scientists whose views align with their own, and discount misaligned voices as less credible, highlighting the need for strategies to reduce polarization while preserving the integrity and reach of scientific discourse.

Additionally, our work contributes to the literature on social identity, originally developed by Tajfel and Turner (2003) and later incorporated into economic analysis by Akerlof and Kranton (2000) and Shayo (2009, 2020). Recent research has examined how social identity shapes economic behavior in areas such as trade policy (Grossman and Helpman, 2021), teamwork (Charness and Chen, 2020), bonus acceptance (Bursztyn et al., 2020a), policy adoption (Garcia-Hombrados et al., 2024), and everyday interactions (Braghieri et al., 2024). Party affiliation increasingly influences individual choices beyond politics, including decisions about marriage (Alford et al., 2011), dating (Huber and Malhotra, 2017), residential location (Brown et al., 2022), career path (Chinoy and Koenen, 2024), medical treatment (Kim, 2024), and even food preferences (Burnitt et al., 2024). Our findings show that scientists' social identity, specifically their political identity, reduces their perceived credibility, the credibility of their research, and the public's willingness to engage with their communication.

While our study is not a real-time Twitter experiment, it relates to recent experimental work on interventions to reduce misinformation, toxicity, and discrimination online (Guriev et al., 2023, Jiménez Durán, 2022, Beknazar-Yuzbashev et al., 2022, Angeli et al., 2022, Ajzenman et al., 2023a,b). We also contribute to the broader literature on the political effects of the internet and social media, including their role in shaping voting behavior, protests, polarization, and misinformation (see Zhuravskaya et al., 2020, for a review). Our focus, however, is on the reputational risks of the growing role of social media in academic communication.

The remainder of the paper is organized as follows. Section 2 examines the political engagement of a large sample of U.S. academics on Twitter. Section 3 presents survey evidence on scientists' own views about political expression online. Section 4 experimen-

tally tests how political expression affects public perceptions of scientists' credibility and engagement with their content. Section 5 concludes.

2. SCIENTISTS' ONLINE POLITICAL EXPRESSION

Given the growing emphasis on academic impact, we assess whether social media is increasingly vital for scientific dissemination. We track the online diffusion of over 100,000 Scopus-indexed articles published from 2011 to 2020 in top general interest and life-science journals using Altmetric data (Alabrese, 2022, Peng et al., 2022).¹¹ Figure 1, Panel A, shows a steady rise in online presence, particularly on Twitter, where over 96% of articles are mentioned. Panel B further shows that the distribution of Twitter mentions has become less skewed towards zero with a thicker right tail and more high-mention outliers over time, indicating increasing variability in social media attention.

These trends highlight Twitter's growing role in scientific communication while also increasing public visibility into scientists' social media activity. Since about half of those who engage with academic content on Twitter are nonacademic (Mohammadi et al., 2018, Maleki and Holmberg, 2023), and perceptions of political bias among scientists contribute to distrust in science (Altenmüller et al., 2024). Recent international evidence further confirms that people frequently encounter and discuss science on social media—and that in some countries, notably the United States, individuals report having participated in protests and activism around scientific issues (Mede et al., 2025).

These patterns suggest the importance of systematically tracking how scientists themselves engage in online political discourse. In this section, we therefore leverage large language models (LLM) to cost-effectively measure the extent and nuance of scientists' political expression on social networks.

2.1. Academics on Twitter: Sample and Method

We analyze political discourse and ideological polarization among 97,737 U.S. academics on Twitter from 2016 to 2022, a substantial subset of the 683,050 U.S. research-active academics (National Center for Education Statistics, 2020). These academics garner

¹¹Of the 114,868 articles published between 2011 and 2020 in *Science*, *Nature*, *PNAS*, *Cell*, *NEJM*, and *Lancet*, 107,008 were tracked by Altmetric as of November 10, 2021. See Appendix Section 3 for details.

significant attention to their online content: over 2016–2022, the number of retweets per post averages 16.4 (SD = 1,251.4; median = 0.3), with 6.7 retweets on average for their politically salient tweets (SD = 134.9; median = 0.4).¹² These university-based researchers were identified by Mongeon et al. (2023) by matching authors’ OpenAlex IDs to Twitter/X accounts via Crossref Event Data. Their method achieved high precision and moderate recall by identifying authors who tweeted their own research.¹³ Building on Garg and Fetzer (2025b), we use their merged dataset of Twitter timelines and OpenAlex records, employing LLMs to detect whether tweets address specific salient topics and to classify stances of posts as pro, anti, or neutral towards those topics.¹⁴

Our dataset comprises approximately 116 million tweets (original tweets, retweets, quote retweets, and replies) from these academics, and our analysis focuses on five politically salient issues: Abortion Rights, Climate Action, Racial Equality, Immigration, and Income Redistribution.¹⁵ The procedure involves two key steps. The first is *topic detection*. OpenAI’s GPT-4 was used to generate dynamic keyword dictionaries designed to capture the evolving discourse on each topic. The model was prompted with:

Provide a list of [ngrams] related to the topic of [topic] in the year [year]. [Twitter Fine Tuning].
Provide the [ngrams] as a comma-separated list.

¹²If we instead examine the total stock of retweets accumulated by each academic during 2016 – 2022, the mean is 1,100 retweets (SD = 6,241; median = 270) overall, and 1,217 retweets (SD = 2,602; median = 414) for politically salient tweets.

¹³Mongeon et al. (2023) validate their matching against 13,208 ORCID–Twitter pairs. They recover 8,229 of those (True Positives), miss 4,979 (False Negatives), and hallucinate 361 extra pairs (False Positives), yielding a precision of 0.958, a recall of 0.623, and an F_1 of 0.755. Put differently, roughly 4% of the handles in their file are false positives, while $\approx 38\%$ of scholars on Twitter remain undetected. Because we analyse the resulting high-precision, moderate-recall subset without adding new accounts, our findings speak to the behaviour of the $\sim 60\%$ of research-active U.S. academics who both maintain a Twitter/X presence and occasionally self-promote their scholarship there. If the remaining non-self-tweeting scholars are, on average, less vocal or less politically engaged online, then any bias in our estimates is likely to be conservative—that is, our measures may slightly overstate how expressive (and polarised) the wider professoriate is on social media.

¹⁴Comprehensive individual-level, time-series data are publicly available from Garg and Fetzer (2025b).

¹⁵We identified salient issue based on Gallup survey. Gallup’s list is available [here](#).

The [*Twitter Fine Tuning*] either explicitly instructed the model to ‘Focus on language, phrases, or hashtags commonly used on Twitter’, ensuring the dictionary remains contextually relevant to the platform’s discourse or was left empty. This process, spanning 5 topics $\times$ 3 n-gram types $\times$ 7 years $\times$ 2 vernacular adjustments (210 prompts), yields five comprehensive keyword dictionaries applied to the full corpus of tweets. Tweets containing any keyword from a topic’s dictionary were labeled as belonging to that topic (e.g., tweets mentioning “Paris Agreement” are labeled under Climate Action). The topic detection captures 5.31% (around 6.15 million) of the initial 116 million tweets. Climate Action was the most discussed topic (2.09% of all tweets), followed by Racial Equality (1.50%) and Immigration (0.86%).¹⁶

The second step is *stance detection*. Using OpenAI’s GPT-3.5 Turbo, topical tweets were classified into three discrete categories: “pro,” “anti,” and “neutral.”¹⁷ The classification used the following prompt:

Classify this tweet’s stance towards [*topic*] as ‘pro’, ‘anti’, ‘neutral’, or ‘unrelated’. Tweet: [*tweet*].

This process not only identifies the position of each tweet on a specified topic but also enhances precision and relevance by filtering out unrelated tweets. To reduce costs, we sampled up to three random tweets per author, month, and topic (99.55% had one tweet; 0.45% had two; 0.004% had three), ensuring adequate coverage and reliable stance estimation. See Appendix Table VI for examples on each topic-stance, posts-level summary statistics in Appendix Table III and details on validation in Appendix Section 4.

¹⁶Using GPT-4 on all tweets would be prohibitive, so we focus on using GPT-3.5 Turbo on the set of topical posts for stance detection.

¹⁷Our “pro/anti/neutral” tags do not seek to code tone (e.g. partisan invective) but rather what social-science work calls a directional policy valence: does the tweet, implicitly or explicitly, point in the policy direction of more (or less) X? A tweet that merely cites evidence (“Our LancetGH paper finds hostile-environment policies harm migrants’ health”) still carries a directional signal—restrictive policy $\rightarrow$ worse outcomes—which readers will reasonably interpret as evidence against such restrictions. Hence it is coded pro-immigration. Likewise, flagging that “99.9% of the literature finds human-caused warming” provides empirical support for mitigation and is therefore pro-climate-action. We argue that this does not exaggerate politicization. Our measures do not require overt partisan language, so it will indeed classify some fact-sharing as political expression. This is by design: our research question is how often academics go on record with a position that has policy implications, regardless of rhetorical style.

Appendix Table II summarizes the key characteristics of our sample of academics. Most scientists are in STEM, followed by Medicine and Social Sciences. 54% of U.S.-based academics discussed at least one topic, and 81% of those expressed non-neutral stances. Our analysis thus focuses on 42,747 scientists who posted non-neutral topical tweets.

We call *politicized* the scientists who express a non-neutral stance on any salient topic. Our focus on non-neutral communication offers several advantages: first, it maps naturally from our methodology; second, it broadly identifies the presence of academic discourse related to political issues. Such discourse may enable the public to infer scientists' political leaning and update the perception of their credibility: a central concern addressed in the second part of this paper.

Appendix Table II further breaks down the proportion of politicized scientists by key characteristics. The proportion remains relatively stable across categories of academic recognition (between 41% and 46%).¹⁸ Disciplinary differences are more notable, with higher proportions in Humanities (58%) and Social Sciences (65%) versus STEM (43%) and Medicine (38%). Some gender differences also emerge, with 50% of female scientists and 40% of male scientists being vocal. More details are provided in Section 2.2.3.

2.2. Academics Political Engagement on Twitter

Figure 3 compares political discourse trends between academics (orange) and a random sample of approximately 100,000 U.S. Twitter users (blue) between 2016 and 2022.¹⁹ The top-left panel ("All") shows that well over one-third of the 97,737 tracked U.S. academics express a non-neutral opinion on any specified salient topic in each month, revealing a substantial gap compared to general users.²⁰

Breaking the trend down by topic, Climate Action and Racial Equality (the most frequently discussed issues) are the largest contributors to the gap. Notably, Climate Action

¹⁸Cumulative citation categories: <100; 101–500; 501–1,000; >1,000. Categories are derived from the following distribution: 25th percentile = 36, median = 190, mean = 1205, 75th percentile = 849.

¹⁹Twitter users were originally sampled by Siegel et al. (2021). Their method randomly generated user IDs and filtered for active accounts with geographic metadata indicating U.S. residency, based on time zone or location fields referencing U.S. states or major cities.

²⁰Monthly aggregated scatter plots are smoothed using LOESS (span=0.5) with shaded standard errors.

discourse declined at the onset of the COVID-19 pandemic, while Racial Equality remained highly salient, surging in mid-2020 following the George Floyd incident and widespread protests. Abortion Rights discussions spiked in 2022 due to changes in abortion laws, and Immigration conversations peaked around the 2016 presidential election and diminished over time.

As our working definition of politically engaged academics is broad, encompassing any scientist expressing a non-neutral stance on a politically salient topic, we conduct further analysis to explore (i) the extent to which such tweets also mention research-related content and (ii) whether their statements are factual or opinion-based. These exercises help distinguish between politicization through interpretive or normative commentary and neutral communication of scientific facts.

2.2.1. *Overlap with Politics and Research*

To better understand the interplay between scientific discourse and political engagement, we analyze how academics reference both scientific research and political figures in their tweets. This allows us to assess whether topical engagement primarily reflects the communication of evidence, political commentary, or both. Following the methodology in Section 2.1, we used GPT-4o to generate dictionaries for detecting mentions of [*scientific research papers*]. Additionally, we generated dictionaries for [*Donald Trump*], [*Joe Biden*], [*Politicians*], and [*Political Candidates*]. These were combined to create a comprehensive dictionary for identifying mentions of political figures (see Appendix Table IV for examples).

We then examined the yearly distribution of academics' tweets across four categories: tweets mentioning politicians (blue), research papers (green), both (pink), and other content (grey). Appendix Figure 1 shows that, on average, 10% of tweets mention politicians, 20% reference research papers, and about 1% mention both. These proportions remain relatively stable over the years, with fluctuations likely tied to significant events (e.g., larger coloured areas in 2020 may reflect discussions post-George Floyd or a COVID-19-induced surge in research tweets). Darker shades mark tweets addressing the five salient issues that largely fall in the residual grey category, namely Abortion Rights, Climate Action, Racial Equality, Immigration, and Income Redistribution.

These findings align with the cross-sectional summary statistics in Appendix Table III: overall, 9.8% of tweets mention politicians and 19.2% mention research papers; when restricted to the five salient topics, these rise to 16.0% and 28.9%, respectively. Notably, Climate Action tweets mention research papers 44.5% of the time versus 15% for politicians (a roughly 3:1 ratio), while Abortion Rights and Immigration tweets favor mentions of politicians (25.6% and 28%) over research papers (15% and 21.4%), with ratios of approximately 1.7:1 and 1.3:1.

Overall, the data suggest that scientists engage in discussions about research, political figures, and politically salient issues, with a moderate overlap between these domains. Given that politically salient tweets generate high engagement and research-related discourse is increasingly prevalent on Twitter (see Figure 1), it is important to examine not only whether scientists engage with political or research content, but also whether their posts rely on factual claims or reflect broader opinion-based discourse.

2.2.2. *Factual vs Non-Factual Content*

We further investigate the nature of topical engagement among scientists by assessing the factual veracity of their tweets. This analysis serves to distinguish between tweets that reflect politicized opinion and those that convey factual content, which may nevertheless align with a political stance. Although our working definition of politicized scientists is based on the expression of a *non-neutral stance* on a salient topic, this measure does not rely on tone or intent. By classifying the veracity of tweet content independently, we provide a complementary lens on the nature of topical engagement, shedding light on whether scientists express positions through opinionated commentary or through the communication of verifiable claims.

We assess whether topical tweets contain factual or non-factual statements using GPT-4o-mini. Tweets are classified into four mutually exclusive categories based on whether they include or imply a claim that is: TRUE, FALSE, UNCLEAR, or NO FACTUAL CONTENT (the default for absence of any checkable claims). The classification follows a structured prompt that also requires the extraction of any factual claims and a justification for the assigned label (see Luu et al., 2024, for a similar approach). The full prompt is provided in Appendix Section 4.

Overall, 51% of tweets are classified as NO FACTUAL CONTENT, indicating that the majority of topical posts consist of opinions or general commentary rather than verifiable claims. Among the 49% of tweets that do contain factual content, 2% are labeled FALSE, 53% TRUE, and 45% UNCLEAR (for example, due to unverifiable forecasts or ambiguous wording).

Clear differences emerge across ideological stances (Figure 2). Tweets classified as *anti* (more conservative) are more likely to be labeled as FALSE or UNCLEAR than either *neutral* or *pro* (more progressive) posts. In contrast, *pro* tweets are most often categorized as NO FACTUAL CONTENT, while also showing the lowest share of FALSE and UNCLEAR classifications. To account for the possibility that retweets may introduce ambiguity (e.g., through critique or irony), we also exclude retweets from the classification. Doing so generally increases the share of NO FACTUAL CONTENT tweets but leaves all other patterns unchanged (available upon request).

This exercise addresses a key concern that some tweets labeled as *pro* or *anti* may not reflect politicized expression, but rather factual reporting that aligns with a given issue (e.g., new evidence on climate change or immigration policy). For example, a tweet can be labeled as "pro-climate" based on its content, while simultaneously being categorized as TRUE under our classification. Our results show that this is not systematically the case: while many stanced tweets (especially in the *pro* category) are grounded in verifiable facts, over half of non-neutral tweets contain no checkable claim at all. The factual classification thus complements the stance measure by distinguishing between opinion, ambiguous claims, and factual reporting within politicized topics.

2.2.3. Differences by Gender and Discipline

We examine variation in academics' online political expression by gender and discipline. Gender was determined via a binary LLM classification (Appendix Figure 2).²¹ Academics with female-labeled names express slightly more political opinions. This is especially true for Abortion Rights, Immigration, and Racial Equality, where activity increases after 2020. For Climate Action and Income Redistribution, there are no significant gender differences.

²¹Names were classified as "Male" or "Female" with 99% accuracy (see Appendix Section 4).

We also assess disciplinary differences using OpenAlex "Concepts." Each publication is scored on 19 root concepts; for each researcher, we average these scores over the entire period and assign the primary concept to one of three groups: Medicine, Social Sciences (Business, Economics, History, Political Science, Psychology, Sociology), or STEM (Biology, Chemistry, Engineering, Environmental Science, Geography, Geology, Materials Science, Mathematics, Physics). Figure 4 shows that, overall, U.S. academics express political views beyond their field. However, STEM scientists are most vocal about Climate Action, while Social Scientists are more active on Income Redistribution. Notably, gaps between disciplines have narrowed over time, contrasting with common stereotypes of scientist politicization (Altenmüller et al., 2024).

2.2.4. Scientists' Ideological Polarization on Twitter

Our analysis thus far shows that, of the 97,737 academics sampled, 52,541 engaged in tweeting about political issues, with 81% of these expressing non-neutral stances on at least one of the five salient topics. We refer to *ideological polarization* to indicate the divergence in political views or issue positions among individuals.

For each topic and user in a given month, we calculated their slant as the difference between the number of pro-stance tweets and anti-stance tweets, divided by the total number of tweets (pro, anti, or neutral) they posted during that month on that topic.²² This is expressed as:

$$S_{um} = \frac{pro_{um} - anti_{um}}{pro_{um} + anti_{um} + neutral_{um}}$$

In this formula, S_{um} denotes the net pro stance share of tweets by user u in time-period m , relative to all their tweets on a topic. This approach provides, for each individual, a continuous measure of slant ranging from -1 (completely anti) to 1 (completely pro) for each topic, at any point in time. Our method captures the spectrum of opinions, from strong opposition to firm support, while distinguishing between more moderate or explicit posi-

²²Specifically, we measured individual academics' slant following the theoretical framework of Esteban and Ray (1994), categorizing tweets into 'pro', 'anti', and a residual 'neutral' category.

tions, a feature essential in environments where public opinions may lean toward socially desirable expressions (Bénabou and Tirole, 2006).

Figure 5 displays the net pro stance distributions S_{um} for each topic across academics and general users over the entire sampled period. Except for Racial Equality, neutral views dominate. However, multiple peaks, confirmed by Hartigan’s Dip Test (Hartigan and Hartigan, 1985) ($p=0$), indicate strong multimodality and distinct political camps. General users cluster more at extreme conservative stances (e.g., -1) than academics, who concentrate in the moderate liberal/progressive range. Notably, for Racial Equality, with few neutral stances, users show higher consensus by clustering more at the fully pro stance compared to academics. Appendix Figure 3 further compares early (2016–2017) and late (2021–2022) net pro stance distributions. Results suggest shifts in opinions over time, particularly on topics like Immigration and Abortion Rights, for which KS is highest (Massey Jr, 1951), highlighting the dynamic nature of ideological polarization among academics.

We also examine differences in net stances between tweets mentioning scientific research or not and how differences in Twitter reach and post type influence academics’ net stances. Appendix Figure 4 shows that academic research-related tweets are significantly more moderate than non-research tweets (except for Climate Action, where differences are smaller and research tweets are slightly more progressive). The largest divergence between research and non-research-related tweets is seen in Abortion Rights: non-research tweets skew more pro-choice (+1) while research tweets cluster more near neutrality (0). Further, Appendix Figure 5 indicates that academics with lower Twitter reach (below the median of 522 followers) exhibit more extreme stances, whereas high-reach academics appear more moderate. Finally, Appendix Figure 6 reveals that retweets display consistently more extreme stances than original tweets, particularly for Abortion Rights and Climate Action, suggesting that original posts tend to be more measured while retweets amplify polarized views.

The pattern highlighted in Section 2.2.2 are also reflected across topics (Appendix Figure 7). For each factual category (TRUE, FALSE, UNCLEAR), the distribution of tweets is generally the same as the general sample just described, indicating that most academic posts, regardless of veracity, cluster around moderate positions. Within the more ideologically extreme regions, however, we observe notable relative differences across factual categories. FALSE tweets are more prominent within conservative-leaning clusters, while

TRUE or NON FACTUAL tweets are relatively more common in progressive-leaning clusters. Racial equality is a marked exception to this pattern. While most tweets on the topic oppose racism across the spectrum, TRUE posts are relatively more concentrated among conservative-leaning tweets. In contrast, FALSE and UNCLEAR posts are more evident among progressive-leaning ones, and NON FACTUAL tweets are somewhat in between.

Findings align with expectations: research-related tweets likely reflect professional norms of moderation, while non-research tweets allow for more direct political expression. Similarly, academics adopt a more measured tone in their original posts, potentially exercising greater caution with content directly attributable to them. Finally, with the exception of Climate Action, opinion-based content (NON FACTUAL tweets) tends to be more ideologically extreme (particularly in a progressive direction) compared to tweets containing verifiable factual statements. Overall, these patterns highlight the sensitivity of our measure in capturing differences between more neutral or explicit positions.

2.2.5. *Evolution of Scientists' Disagreement*

We finally examine the evolution of disagreement among scientists by computing the monthly variance of net pro stances S_{um} for each topic. Variance provides a continuous measure of opinion diversity that is insensitive to the average stance yet highly sensitive to extremes, facilitating longitudinal and cross-topic comparisons (McCarty et al., 2016). Although variance is robust and interpretable for large datasets, it does not reveal the causes of disagreement (Fiorina et al., 2005) and may overstate divergence if few hold extreme views; it also may not capture consensus well (Hopkins, 2018). Therefore, we interpret variance in relative terms—lower variances indicate greater consensus—and examine its fluctuations over time and topics to identify both divergence and emerging agreement in the evolving ideological landscape.

The top left panel in Figure 6 ("All") aggregates all topics, showing a mild increase in disagreement from 2016 to 2022, which could raise concerns about scientific consensus and public trust. While scientists appear to exhibit greater ideological distance than general users overall, a topic breakdown reveals nuances. For most topics, the general U.S. Twitter population shows higher disagreement than academics. However, for Racial Equality, academics display a growing and more pronounced disagreement compared to the public—a disparity that disproportionately affects the aggregate trend given its high prominence. For

Immigration, Income Redistribution, and Abortion, public disagreement is higher, with the gap widening during the pandemic before narrowing toward the end of the period. In the case of Climate Action, disagreement between academics and the general public persists throughout, with the gap widening further by the end of 2022.

3. SCIENTISTS' VIEWS ON ONLINE POLITICAL EXPRESSION

The most immediate advantage of social media for scientists is access to audiences beyond their immediate academic networks. Appendix Figure 8 (left panel) shows that academics who were already active on Twitter in 2016 became significantly more visible over time, as measured by average retweets per monthly post in 2016 (blue) versus 2022 (orange), based on their initial posting activity. This pattern is even more pronounced among those who occasionally engaged with political topics (right panel), who received higher engagement than their peers, both at the beginning and at the end of the period.

This visibility, however, may come at a cost. As scientists become more public-facing, particularly on politically sensitive topics, they risk being perceived as biased or partisan. Such perceptions can erode their most valuable asset: credibility. The remainder of this section investigates whether scientists anticipate this trade-off and how they evaluate the professional costs and benefits of political expression online.

To address these questions, we examine scientists' self-reported views using two surveys. We first recruited 128 scientists worldwide via Prolific.²³ Appendix Table VII describes the sample: 94% are employed (over 60% in universities), including 43% postdoctoral researchers, 28% faculty, and 29% industry professionals; 34% work in life sciences/biomedicine, 34% in social sciences, and the rest in physical sciences or technology.

First, we quantified whether scientists expect a credibility penalty for expressing political opinions on social media—that is, whether they effectively anticipate the reputational cost that our core experiment in Section 4 uncovers. Second, following Bursztyn and Yang (2022), we elicited both the scientists' own views and their perceptions of the opinions of peers, that is, their first- and second-order beliefs on the acceptability of political expression within and beyond their expertise. Third, we asked about personal experiences with publicly expressing political opinions.

²³Prolific respondents were screened for employment in "Research," PhD-level education, and English fluency.

To gauge the expected credibility penalty, we told respondents that scientists who **do not** express political opinions receive a trust score of 7.2/10 based on our main experiment (see next Section 4), then asked: "*What is the reported trust level for scientists who **do** express political opinions on social media?*" (participants earned a £0.50 bonus for a correct response). As shown in Figure 7, respondents expect a 30% loss in public trust, nearly double the experimentally observed loss of 16.6%.

While the perception of a reputational loss aligns with cheap talk models under reputational concerns (Morris, 2001, Ottaviani and Sørensen, 2006), the overestimation of the penalty stands in contrast to these models, which assume that reputational costs are common knowledge and that beliefs are correct in equilibrium. Exploring this discrepancy is beyond the scope of this paper, but we offer two possible explanations. One is that a more prominent audience for scientists' reputational concerns may be fellow scientists rather than the U.S. public or journalists, as in our experiment. Another is that scientists may simply misperceive the beliefs of the U.S. public, consistent with findings in Bursztyn et al. (2020b).

Relative to professional norms, Panel A of Appendix Figure 9 shows that scientists consider political expression acceptable when related to their expertise, but not beyond it.²⁴ Panel B shows they believe peers share this norm. Finally, Panel A of Appendix Figure 10 indicates that scientists are significantly more hesitant to comment on topics outside their field,²⁵ while Panel B shows they expect net negative repercussions for out-of-field comments and net benefits for in-field ones, though these net differences are small.

To further examine the perceived professional costs and benefits of expressing political views online, we again surveyed 118 international scientists via Prolific. This survey disentangles the perceived impact of political expression on specific professional outcomes such as network size, citations, academic job offers, media appearances, policy roles, and top publications.

Figure 8 compares scientists' perceived outcomes for three hypothetical profiles: Scientist X (grey), who expresses political views related to their research; Scientist Z (red), who

²⁴This aligns with Garg and Fetzer (2025b), who find that academics adopt more pro-social views on topics where they hold expertise.

²⁵This complements evidence of a 'spiral of silence' in academia (e.g., Norris, 2020).

expresses views on unrelated topics; and Scientist Y, identical to either X or Z but refraining from political expression. Outcome distributions shifted toward X or Z (green areas) indicate perceived benefits, while a denser left tail for Y (blue areas) signals perceived costs. The results reveal clear perceived benefits of political expression for media appearances and policy roles, milder benefits for network size and citations, and ambiguous or slight costs for top publications and academic job offers.

Finally, we posed three open-ended questions to capture scientists' views on what constitutes "politicized" behavior, the boundaries of expertise for political commentary, and personal experiences with its costs and benefits. Appendix Table VIII summarizes these responses, highlighting recurring concerns about credibility, expertise boundaries, and the trade-offs of political engagement. While our Twitter data cannot fully capture scientists' motivations, survey responses consistently suggest that academics perceive greater benefits when political opinions align with their research expertise.

4. SCIENTISTS' POLITICAL EXPRESSION AND PUBLIC PERCEPTIONS

Having examined the scope of online political discourse and issue-based disagreement among scientists, as well as scientists' perceptions of costs and benefits of political expression, we now assess the risks of such expression.²⁶

Specifically, we study how scientists' political expression affects public perceptions of their credibility and contributes to audience polarization. We ran a conjoint experiment with 1,704 respondents from a broadly representative U.S. sample, recruited via Prolific: a platform widely used for experimental research (Bursztyn et al., 2023a, Enke et al., 2023).

Our sample broadly reflects the U.S. population in terms of key dimensions, including political affiliation, region, ethnicity, and gender. As is typical with Prolific, respondents tend to be slightly younger, more educated, and have higher incomes than the general population. Importantly, prior work shows that Prolific's Republican sample is representative of the broader Republican population (Braghieri et al., 2024, Kashner and Stalinski, 2024). Appendix Table IX provides full demographic comparisons with the population census. To

²⁶An overview of all experiments and surveys conducted for this paper can be found in Appendix Table I. All studies and surveys were pre-registered on [AsPredicted](#) with numbers [166935](#), [179009](#), [181452](#), [186295](#), [206927](#), and [235820](#).

ensure data quality, all participants passed an attention check before beginning the main task.

We opted for an online experiment over a field experiment on X due to several advantages, despite a minor conceptual mismatch with Section 2. First, the conjoint design on Prolific lets us systematically vary scientists' attributes—something hard to achieve on X. Second, a field experiment would require creating fake scientist profiles to engage with users, a method both impractical and likely to raise authenticity concerns. Third, typical field outcomes like follow-backs and retweets (Ajzenman et al., 2023a,b) are too narrow for capturing perceived credibility.

4.1. Experimental Design

Following best practices for conjoint experiments (Hainmueller et al., 2015), we created five hypothetical scientist profiles, randomly varying key attributes: gender (male/female), research field (Social Sciences, STEM, Medicine, Humanities), seniority (junior/senior), university prestige (high/low), and political affiliation.

The latter (our main attribute) is conveyed through a Twitter-style bio and a high-engagement tweet written by real scientists,²⁷ spanning five categories: Strong Democrat, Moderate Democrat, Neutral (baseline), Moderate Republican, and Strong Republican.

The five political profiles were created by randomizing the remaining attributes, ensuring each had a unique combination of characteristics. Vignettes followed a consistent format and were shown in random order to avoid order effects (see Appendix Table XI and Figure 11 for a visual representation).

The profile you are seeing is a [Gender] scientist.

This scientist works in the field of [Research Field]

Currently, this scientist is a [Seniority] at the [University Affiliation].

The scientist is active on X (formerly known as Twitter).

The Twitter bio of the scientist is: "[Twitter Bio]".

²⁷High-engagement tweets (those with many likes, retweets, and comments) are used for their greater visibility.

Minor modifications were made to the wording of the posts to preserve the anonymity of the academic author.

A recent selected Tweet reads: "[Twitter Post]".

Participants rated the credibility of the scientist and their research separately on a 0–10 scale, and indicated their willingness to read an opinion piece by a similar scientist (0–10).²⁸ To encourage truthful responses, participants were told they would receive an article from a real scientist matching their ratings.²⁹ The design included 160 unique profiles, generated from all combinations of 2 (gender) $\times$ 4 (field) $\times$ 2 (seniority) $\times$ 2 (university affiliation) $\times$ 5 (political profile).

Our within-subject conjoint design covers the full political spectrum and provides greater power to detect the effect of each attribute, even though the tweets are not identical in content and style. A between-subject design, by contrast, would allow only limited variation: for example, varying the Twitter bio while holding the tweet constant. However, to rule out any potential confounder, we also ran a between-subject task in which respondents rated the credibility of an economist (discipline held constant). In this additional task, detailed in Section 4.4, the tweet (promoting a research paper) was fixed, while political leaning was signaled only through the bio (left or right). This design allowed us to isolate whether credibility penalties stem from politically salient research topics or from identity cues unrelated to scientific content.

Our design addresses common concerns in conjoint experiments (Hainmueller et al., 2015), particularly attribute-order effects and experimenter demand effects. To mitigate order effects, we placed the primary attribute—political affiliation—at the bottom of the profile. To reduce demand effects, we highlighted multiple salient attributes and informed respondents that they would receive an opinion piece from a real scientist matching their top-rated profile. We explicitly discuss and bound potential demand effects by conducting a robustness experiment where we explicitly induce demand effects, as discussed in Sec-

²⁸Hainmueller et al. (2015) shows that single- and paired-vignette designs yield similar estimates; we use the simpler single-vignette approach.

²⁹The same incentivization that we employ has been widely used in experiments in labor economics (Kessler et al., 2019, Cullen et al., 2023, Kessler et al., 2024), gender economics (Low, 2014), finance decision in development countries (Macchi, 2023, Colonnelli et al., 2024), and in education (Exley et al., 2024). While the elicitation procedure might not be fully incentive compatible, prior work finds no significant difference between hypothetical and incentivized responses (Hainmueller et al., 2015, Brañas-Garza et al., 2021, 2023, Enke et al., 2022).

tion 4.2.2. To enhance profile plausibility, we informed participants that all tweets were from real scientists and stated in the consent form that no false information was used. Full experimental instructions are provided in the supplementary [Instructions Material](#).

4.1.1. *Validating Twitter Political Signals*

To validate our political affiliation signals, we surveyed 98 new Prolific participants who categorized each combination of Twitter bios and tweets from the main experiment into one of five labels: Strong Republican, Moderate Republican, Neutral, Moderate Democrat, or Strong Democrat. Appendix Figure 12 shows that most responses matched the intended labels, with misclassifications largely limited to adjacent categories. This high accuracy confirms that our signals were reliably perceived, supporting the credibility of our design.

4.2. *Impact on Public Perceptions*

Figure 9 shows that scientists' online political expression leads to a monotonic penalty: neutral profiles receive the highest ratings for personal credibility, research credibility, and willingness to read opinion pieces, with outcomes declining significantly as political affiliations become more extreme on either the left or right.

Scientists expressing any political stance are viewed as less credible than neutral ones (Panel A), supporting the stereotype that scientists should remain impartial ([Altenmüller et al., 2024](#)). Relative to neutral scientists, Strong Republicans are rated 39% less credible and Strong Democrats 11% less. Moderate Republicans and Moderate Democrats incur smaller penalties: 9% and 7%, respectively. An identical pattern holds for research credibility, indicating that political discourse impacts both personal and scientific credibility.

Similarly, willingness to read declines monotonically: respondents are 42% less willing to read from Strong Republicans and 10.7% less willing to read from Strong Democrats. Moderate Republicans and Moderate Democrats see more limited declines of 8% and 4.5%, respectively. All results remain robust when controlling for respondent characteristics (Appendix Figure 13, Panels C and D).

A model of conformity like [Bernheim \(1994\)](#) proposes a mechanism that rationalizes why the scientists who engage in political communication experience a credibility penalty. In [Bernheim \(1994\)](#)'s model of conformity, individuals care directly about social status, which depends on public perceptions of their underlying predispositions, and small depar-

tures from a homogeneous norm incur smoothly increasing but endogenous status penalties. Interpreting scientific neutrality as such a norm, the expression of partisan stances constitutes a departure that signals bias and thus lowers status.

While political affiliation is the most salient attribute, a few additional characteristics also marginally affect public perceptions. Scientists from prestigious institutions (e.g., Harvard, UC Berkeley, UChicago) are rated as more credible and attract higher readership than those from less prestigious ones (e.g., University of Arkansas, University of Connecticut), as shown in Appendix Figure 13 and Appendix Tables XII to XIV. When examining each political profile separately, the benefit of a prestigious institutional affiliation holds for all except the neutral academic, and it even results in a penalty for Strong Democrat profiles. Moreover, Full Professors are generally seen as more credible and are more likely to be read than Assistant Professors.

Other attributes are never statistically significant. Specifically, gender does not appear to affect credibility—though neutral male scientists are (non-significantly but consistently) less likely to be read than their female counterparts, partly consistent with findings in Sievertsen and Smith (2025).

4.2.1. *Heterogeneity by Respondent Leaning*

To determine whether scientists' online political expression may contribute to polarizing audience views, we analyzed how perceptions of different scientist profiles vary based on respondents' political leanings. Panel B of Figure 9 reveals significant heterogeneity tied to political identity.

Respondents identifying as *Democrat* or *leaning Democrat* assign substantially lower credibility to the Republican-leaning scientists: the Moderate Republican scientist faces an 18.3% credibility penalty and the Strong Republican a 60% penalty (Panel B, dark blue), with willingness-to-read dropping by 23.2% and 69%, respectively (light blue). These respondents view the Neutral and Democrat-leaning scientists similarly.

Conversely, respondents identifying as *Republican* or *leaning Republican* rate the Democrat-leaning scientists as 16% (Moderate) to 26% (Strong) less credible (Panel B, dark red), and are 18%–30% less willing to read their opinions (light red). Notably, they view the Moderate Republican scientist as 3.1% more credible than the Neutral one and are 11.6% more willing to read them. While the Strong Republican scientist is rated 12% lower

in credibility than the Neutral one, they remain more credible than the Democrat-leaning scientists, with only a 7% drop in willingness-to-read.

Overall, the gap in willingness to engage with scientists holding opposing political identities ranges from 18%–23% for moderate stances and increases to 30%–69% for extreme ones. These results, consistent with [Ajzenman et al. \(2023b\)](#) and [Rathje et al. \(2021\)](#), show a monotonic pattern: stronger political signals lead to larger penalties among politically opposed respondents. Crucially, the drop in personal credibility extends to perceived research credibility and translates into lower willingness to engage with their content. [Barrios et al. \(2024\)](#) documents a comparable effect: a 30% decline in trust in an economics paper due to a conflict of interest, aligning closely with our findings.

Although the scientists’ tweets themselves contain no content that would undermine their expertise or research, the starkly different reactions by Democrats versus Republicans point to in-group/out-group dynamics—where group identity drives belief updating independent of informational value ([Tajfel et al., 1971](#), [Tajfel and Turner, 2003](#))—as the engine of affective polarization.³⁰ To capture this formally, we draw on [Orzach \(2024\)](#)’s dynamic conformity model, in which each observer trades off learning from a sender’s private signal against a “conformity” cost that rises with the squared distance between the sender’s public stance and the observer’s own in-group norm. In our context, Democrats treat neutrality or mild Democratic cues as the normative benchmark, and Republicans treat neutrality or mild Republican cues likewise; any stronger partisan expression thus incurs a conformity penalty. Once the conformity cost lies above the threshold for pooling equilibria, observers ignore private information entirely and “pool” on their in-group norm, resulting in monotonically larger discounts in both credibility and willingness-to-read as scientists’ signals deviate further from that norm. This mechanism exactly reproduces the Panel B heterogeneity: strong out-group signals (e.g., Strong Republicans for Democratic respondents) trigger the largest penalties, moderate out-group signals incur intermediate penalties, and neutral or in-group signals face little or no discount.

³⁰In-group favoritism enhances positive affect toward one’s own group while fostering disdain for perceived out-groups, and polarization intensifies when group identities become central to self-concept ([Iyengar and Westwood, 2015](#), [Mason, 2018](#)).

4.2.2. Robustness Checks

We conducted several robustness checks to address potential concerns: (1) sample and procedure robustness, (2) variation in perceived expertise, (3) differential respondent attention, (4) misperception of political profiles, (5) multiple hypothesis testing, and (6) experimenter demand effects.

Replication among Twitter users. A potential concern is that the credibility penalties observed in our general-population samples might differ among active Twitter users, who are more accustomed to political content online. We therefore ran a replication with 2,000 U.S. respondents recruited on Prolific who reported using Twitter. Of these, 85% had used Twitter at least once in the past week and 36% used it daily, indicating high platform engagement.³¹ Appendix Figure 20 shows that the estimates are virtually unchanged relative to those in the main experiment with a general-population sample, confirming that our core findings generalize to a population more accustomed to political communication on social media.

Replication with Modified Outcomes We also replicated the main experiment with 2,000 new Prolific respondents, eliciting only two outcomes: credibility and willingness to read. Results are virtually identical, confirming robustness to procedural changes and to a new, representative U.S. sample (available upon request).

Role of Expertise As discipline is randomized across profiles, we test whether matching a scientist's expertise to the topic of their post affects perceptions (e.g., the Strong Republican may be a medical expert; the Moderate Democrat a STEM expert). Appendix Tables XII–XIV show no significant effect of such alignment on credibility or willingness to engage.

Carryover Effects A crucial assumption for our design is the absence of carryover effects across profiles. Such effects would imply that a respondent's evaluation of one profile

³¹Because it is difficult to define representativeness for Twitter users, we set quotas for gender and political affiliation to mirror the U.S. population. As shown in Appendix Table X, the resulting sample is younger and more educated than the U.S. average—patterns consistent with a social media user base.

is influenced by those seen before or after, introducing bias. Although unlikely due to randomization, we explicitly test this by analyzing responses within each round. Appendix Figure 16 confirms that results are consistent across rounds, ruling out order effects.

Excluding ‘Speeders’ A common concern in online experiments is that some respondents rush through tasks without paying adequate attention, introducing noise. We therefore exclude respondents who completed the survey in under one minute (median time: 7.2 minutes). Appendix Figure 17 shows results remain stable, indicating inattentive respondents are not driving our findings.

Permutation Test We further test whether the baseline ‘neutral’ scientist inadvertently signals political identity. We randomly reassigned labels across the five profiles seen by each respondent and re-estimated our main result, repeating the procedure 100 times. Appendix Figure 18 shows the resulting coefficients cluster around zero, supporting the neutrality of the baseline.

Multiple Hypothesis Testing One advantage of a conjoint experiment is the ability to simultaneously test multiple attributes, though this requires correction for multiple hypothesis testing. We adjust for multiple comparisons using the false discovery rate method (Benjamini et al., 2006), correcting for 11 treatments (16 attribute categories minus five baselines). Appendix Table XV shows that estimates remain statistically significant after correction.

Experimenter Demand Our design minimizes demand effects by placing political affiliation last among randomized attributes and incentivizing truthful responses via a tailored newsletter. Yet, we bound potential residual effects by conducting a separate experiment with 354 U.S. respondents. After rating a neutral profile, subjects were shown either a Strong Republican or Strong Democrat profile. Before rating the second, they were nudged to rate it either higher or lower to induce demand (de Quidt et al., 2018). Appendix Figure 14 shows that these nudges had little effect: neutral scientists remain the most credible and readable, while left- and right-leaning profiles incur penalties.

4.3. *Impact on Journalists Perceptions*

To assess external validity, we conducted a simplified experiment with 135 international journalists recruited via Prolific. The sample includes experienced professionals employed in various roles and outlets (50% with over 5 years of experience, 78% as reporters or editors, and over half based in the U.S. or the UK), with roughly 60% affiliated with politically oriented outlets (see detailed summary statistics in Appendix Table XVI).

In this streamlined experiment, journalists viewed three scientist profiles—signaling Strong Republican, Strong Democrat, and Neutral ideologies via Twitter biographies (without tweets).³² They evaluated the credibility of the scientists and their research and indicated their willingness to feature an opinion piece from a scientist with such characteristics. To incentivize thoughtful responses, journalists were informed that an opinion piece from their highest-rated scientist would appear in a newsletter shown to 100 readers, with a monetary bonus awarded if their selected piece ranked among the top five preferred by readers.

Appendix Figure 15 replicates our main findings in this new sample of international journalists: neutral scientists are rated highest, while ideologically non-neutral profiles incur significant penalties. Specifically, journalists perceive the Strong Republican scientist and their research as 33% and 32% less credible, respectively, and are 40% less willing to feature their opinion piece, relative to a neutral scientist (Panel A). By contrast, the Strong Democrat scientist is rated 5% less credible (4% for their research) and elicits a 2% lower willingness to feature their opinion piece; however, these smaller effects are not statistically significant. The more pronounced effects for Republican profiles are mainly driven by the liberal majority among journalists (62%): among liberal respondents, penalties for Republican scientists range from 42% to 56% (Panel B), while conservative journalists impose penalties of 18% to 28% on Democrat scientists.

We further explored factors shaping journalists' preferences by surveying their views on disclosing scientists' political leanings (when known), expectations of reader reactions to featuring politically engaged scientists, and willingness to engage with such scientists who are politically active on social media. Appendix Table XVII shows that over half of the journalists believe a scientist's political leaning should be disclosed and that featuring politically active scientists can affect a newspaper's credibility. While they anticipate mixed

³²We reduced the number of profiles to account for the limited sample size while maintaining statistical power.

reader reactions, most remain inclined to reach out to politically active scientists, suggesting that journalists may play a role in amplifying scientists' political opinions and shaping public perceptions of scientific credibility.

4.4. *Separating the Effect of Tweeting about Research on Salient Topics from Signaling Political Identity*

Scientists may share political views while discussing their research on politically charged issues or when expressing opinions unrelated to their work. We aim to disentangle whether these two types of communication have different effects on public perception. Specifically, we test whether perceived credibility is influenced by (1) tweeting about research on politically salient topics, which may signal ideological bias (Altenmüller et al., 2024), or (2) signaling political identity through content unrelated to research (Ajzenman et al., 2023b, Rathje et al., 2021, Levy, 2021). While other mechanisms may matter, our design isolates these two core, policy-relevant channels.

Experimental Design To disentangle the two channels, we added a final task immediately after the main experiment in Section 4.1. Respondents were randomly assigned to one of four conditions that varied whether an economist (who regularly discusses politically salient issues) tweets about recent research on a politically sensitive topic (vs. a non-sensitive one), and whether their Twitter bio includes a clear political signal (vs. a neutral one). Expertise is held constant across all profiles.

In the *passive control*, the economist discusses game theory research on a non-political issue. In the *active control*, the economist tweets about research on the negative impact of policies on migrants' health: a topic with a potentially left-leaning narrative.³³ The difference between active and passive control isolates the effect of politically sensitive research content on perceived ideological bias. Both profiles feature neutral Twitter bios. Two treatment arms add a political signal in the Twitter bio—either *Left* (“advocate for equality”) or *Right* (“proud patriot”)—while keeping the same politically salient research as in the active control (see Appendix Figure 19). This isolates the effect of explicit political identity.

³³The research topic was chosen to clearly align with one political side and oppose the other, facilitating interpretation. Symmetrical effects are expected with right-leaning aligned research.

After viewing one of the four profiles, respondents rated the economist's personal and research credibility and their willingness to read an opinion piece from a similar economist. They were also invited to subscribe to a free newsletter on U.S. socio-economic issues featuring similar economists, delivered via Prolific message (see screenshot in [Instructions Material](#)). Finally, we measured respondents' general trust in scientists using three Likert items, averaged into a composite index.

Results Figure 10 shows that both channels, research content and political identity cues, influence audience perceptions, with effects varying by respondents' political leanings. Among Democrats, perceptions remain stable when the economist discusses non-political research content. However, politically aligned research increases perceived credibility by 8.6% (economist) and 6.3% (research), and raises willingness to read by 33%. Adding a left-leaning bio further increases credibility by 1.7% and willingness to read by 7.1%. By contrast, a right-leaning bio reduces credibility by 8.6%, bringing it back to the non-political baseline, and lowering willingness to read by 4.5%. Republican respondents show the opposite pattern: when the economist discusses politically misaligned research, credibility drops by 12% relative to the non-salient benchmark. This penalty increases by 5% with a left-leaning signal and is reduced by 5% with a right-leaning one. Willingness to read follows a similar trajectory.

Newsletter sign-ups and overall trust in science show similar, though less pronounced, patterns. Correlating newsletter demand and willingness to read validates the latter outcome ($\beta = 0.064$, $p < 0.001$). Recontacting respondents a few weeks later further confirms engagement: of 595 participants who expressed interest in the newsletter, 440 were successfully recontacted, and 86% clicked the newsletter link to join.

The bottom of Figure 10 presents average outcomes for treated respondents relative to either the active or passive control, allowing a clearer comparison between in-group and out-group conditions. In-group respondents (those whose political leanings match the political signal) rate scientists more favorably across all outcomes. Regression results in Appendix Tables XVIII and XIX confirm these findings.

These results highlight two main insights. First, audience responses vary sharply by political alignment: for the same content, credibility and engagement increase when the po-

litical signal aligns with the respondent's views—clear evidence of affective polarization. Second, both channels independently shape perceptions of scientists' credibility.

4.5. *Isolating the Effect of Strategic Considerations behind Scientists' Political Expression*

Our descriptive and experimental evidence establishes a credibility penalty for scientists who express political views, but it does not identify *why* audiences penalize them. In sender–receiver settings, observers can attribute a message either to sincere beliefs or to *strategic* motives (e.g., visibility or audience-building); the attribution shapes how they update about the sender's type.

Guided by this logic, and by our survey evidence that scientists perceive visibility as a salient benefit of online communication, we designed an additional experiment to isolate whether perceived *visibility incentives* drive part of the credibility loss. The key idea is to shift perceptions of strategic intent without altering the message content or the scientist's identity. We implement this in the replication sample of Twitter users, where exposure to political content is more common, and evaluate how making visibility incentives explicit shifts credibility and related outcomes.

4.5.1. *Experimental Design*

In the second task of the Twitter-user replication we specifically implemented a 2×3 between-subjects design to test whether credibility penalties arise because audiences interpret scientists' political expression as *strategically motivated*, adapting [Bursztyn et al. \(2023a\)](#) to academics' political expression.

We varied the slant of the tweet shared by an economist with mid-sized reach whose expertise was described as being in immigration. All tweets reported factual statistics but were framed to suggest three ideological leanings: (i) neutral, (ii) left-leaning, or (iii) right-leaning. Crossed with this, we randomized whether respondents were told *before* viewing the tweet that the economist had been informed *ex ante* that we (the researchers) would promote the post on X/Twitter to increase its visibility ("Promotion"), or received no such

information ("No Promotion"). This manipulation directly links political posting to perceived strategic motives.³⁴

Consistent with the survey evidence in Section 3, the *Promotion* manipulation made salient that the economist's behavior could be interpreted as visibility-seeking, potentially altering respondents' evaluations of the scientist. Specifically, outcomes included: (i) credibility of the economist (0–10), (ii) credibility of the economist's research (0–10), (iii) willingness to read a related opinion piece (0–10), (iv) perceived political bias of the economist (–5 = strongly left to +5 = strongly right), and (v) willingness to subscribe to a newsletter of similar economists (Yes/No). In addition, we elicited second-order beliefs about perceived reputational costs: respondents estimated how many out of 100 Twitter users would rate the economist as highly credible, somewhat credible, neutral, somewhat not credible, or not at all credible.

We preregistered three possible patterns to identify the operative mechanism:

1. *No promotion effect*. If promotion has no effect across promotion conditions, respondents already account for strategic motives when evaluating political expression.
2. *Motivated reasoning*. If promotion *selectively* affects credibility—reducing it when the tweet's leaning is misaligned with respondents' partisanship and increasing it when aligned—this indicates partisan interpretation of strategic motives.
3. *Opportunism*. If promotion reduces credibility *uniformly* across leanings, this suggests a general aversion to perceived opportunism.

Several design choices deserve mention. To balance realism and ethics, we asked three economists to post the tweets on Twitter and then promptly deleted them after promotion, while ensuring that their identities were never revealed to respondents. The tweets

³⁴In the *Promotion* condition, subjects read: "You are matched with a (migration expert) economist who posted the following factual tweet on their personal social media account. Before writing and sharing the post, the economist was informed that we (the researchers) would promote the post to increase its visibility and reach. Promoted posts on X/Twitter are displayed to a broader audience beyond the economist's current followers. Visibility typically extends to users with similar interests (such as other academics, policy professionals, or journalists) based on engagement patterns and topical relevance. Tweet: ...".

In the *No Promotion* condition, respondents read only: "You are matched with a (migration expert) economist who posted the following factual tweet on their personal social media account. Tweet: ...", with all sentences about promotion omitted. Further details in the [Instructions Material](#).

contained only *verifiable factual information*, so that participants were not exposed to misinformation; the wording differed only in framing, and respondents were explicitly told that the content was truthful.³⁵ We omitted explicit source references in the survey to avoid confounding reactions, but separately elicited respondents' judgments of the *credibility of the information* itself.

4.5.2. Results

We report four main findings: (i) the slant manipulation worked and promotion did not change perceived ideology; (ii) promotion alters perceived credibility depending on partisan alignment; (iii) the Trust in Science Index shifts in the same direction; and (iv) second-order beliefs about how others would judge the scientist show no systematic change under promotion.

Manipulation check. We first verify that the *slant* manipulation worked as intended. Respondents who viewed the left- (right-) leaning tweet rated the economist as more left- (right-) biased than those who viewed the neutral tweet. Importantly, adding the information on *Promotion* did not shift perceived ideological bias, indicating that the promotion treatment does not confound the slant manipulation (see Bias columns in Figure 11 and Appendix Table XX).

Promotion effects on credibility. We next test whether making *Promotion* salient (i.e., explicitly informing respondents that the economist had been told *before* posting that *we, the researchers* would promote their tweet) alters perceptions of the scientist. Overall, we find no evidence of a uniform penalty from promotion across conditions, providing no support for the opportunism hypothesis. Effects hinge on partisan alignment. Among Democrats, the promotion information magnifies partisan reactions: relative to the corresponding non-promotion cells, the scientist's perceived credibility is higher when the tweet is left-leaning and lower when it is right-leaning. Neutral promotion is essentially null among Democrats. Among Republicans, the promotion information raises the scientist's perceived credibility for right-leaning tweets and attenuates the penalty for left-leaning

³⁵See the exact posts in Appendix Figure 21. The links to the three facts used to portray a neutral narrative versus a left- or right-leaning one can be found here: [neutral](#), [left-wing](#), and [right-wing](#).

tweets; by contrast, making promotion salient for the neutral tweet is associated with a credibility penalty. Overall, Democrats' responses are consistent with motivated reasoning, whereas Republicans' responses contradict a generalized opportunism story and suggest selective approval of promotion for ideological messages but a dislike of promotion when the message is neutral (see Credibility columns in Figure 11 and Appendix Table XX).

Promotion effects on trust in science. Making *Promotion* salient also shifts the Trust in Science Index, but asymmetrically across respondents' political leanings. Among Democrats, general trust increases when the economist posts a left-leaning tweet under promotion, with a smaller positive effect for the neutral message; promotion of right-leaning tweets has essentially no effect. Among Republicans, promotion increases general trust for both left- and right-leaning tweets, while neutral promotion has no impact (see Trust in Science Idx columns in Figure 11 and Appendix Table XX). These patterns suggest that making our intent to promote the message salient can be read by some audiences as constructive scientific engagement, broadly so among Republicans, and selectively (when aligned) among Democrats, consistent with evidence on divergent normative views of scientists' roles in society and policymaking (Cologna et al., 2025, Capozza et al., 2025).

Second-order beliefs about reputational costs. We also tested whether making *Promotion* salient changes respondents' *second-order* beliefs about how others would judge the economist. After viewing the tweet, respondents allocated 100 hypothetical Twitter users across credibility categories; we analyze the expected share who would find the economist "not credible." Across slant and party cells, differences between promotion and no-promotion are small and imprecise, with confidence intervals overlapping zero (Appendix Table XX columns "Other find Not Credible"). We therefore find no systematic evidence that the promotion information increases the perceived reputational costs of posting, alleviating concerns that our credibility effects are confounded by beliefs about how others would react.

In summary, this second task isolates the role of strategic considerations by making the scientists' *visibility incentive* to promote their tweets explicit. Receivers update on this information, but not uniformly. Among Democrats, the promotion information amplifies the main-experiment pattern, raising the credibility gap between authors of aligned and misaligned messages, consistent with a selective attribution of penalty to strategic motives for

1 out-group signals. Among Republicans, the promotion information closes the credibility 1
2 gap from the main experiment: it boosts credibility for authors of aligned messages and 2
3 attenuates the penalty for authors of misaligned ones, contradicting a generalized “opportu- 3
4 tunism tax.” In parallel, trust in science moves positively under promotion (aligned among 4
5 Democrats; broadly among Republicans). Overall, audiences understand and incorporate 5
6 scientists’ incentives, but updating depends on partisan congruence and normative expecta- 6
7 tions about scientists’ public engagement rather than a universal penalty for strategic 7
8 behavior. 8

9 5. CONCLUSION 9

10
11 This study demonstrates that scientists’ public expression of political views on social 11
12 media significantly shapes public perceptions of their credibility. 12

13 First, we show that political discourse is common among U.S. academics: between 2016 13
14 and 2022, roughly 44% of 97,737 scientists actively discussed political issues on Twit- 14
15 ter—over six times the rate observed among a random sample of U.S. Twitter users. Topics 15
16 like climate change and racial equality are frequently discussed, often with divergent view- 16
17 points. While some of these posts convey verifiable facts, a majority contain no checkable 17
18 claims and are best characterized as opinion-based commentary. This distinction is impor- 18
19 tant for understanding how such discourse may be perceived by the public and contribute 19
20 to assessments of scientists’ credibility. 20

21 Second, using conjoint experiments with 6,000 U.S. representative respondents and 135 21
22 international journalists rating synthetic academic profiles with varied political affiliations, 22
23 we identify a monotonic credibility penalty for scientists who express political views. The 23
24 more extreme the position, on either left or right, the less credible the scientist and their re- 24
25 search are perceived, and the lower the willingness to engage with their content, especially 25
26 among ideologically opposed respondents. A follow-on mechanism test indicates that per- 26
27 ceived *visibility incentives* help explain the credibility penalty. 27

28 Complementary surveys of 128 international scientists reveal that they expect an even 28
29 larger credibility penalty than what we observe experimentally. Scientists broadly view po- 29
30 litical expression as acceptable when related to their expertise, but not beyond it—reflecting 30
31 a prevailing norm within academia. At the same time, they recognize potential professional 31
32 benefits from political engagement, particularly when aligned with their research. 32

Our results highlight a significant challenge to the Mertonian norm of Universalism (Merton, 1973), which holds that scientific work should be judged on its merits, not on the scientist's identity or beliefs. While a comprehensive welfare analysis is beyond our scope, the findings highlight an important trade-off between visibility and credibility, with clear policy implications. It may be disheartening yet unavoidable that political opponents react negatively to research tweets on politically charged topics, but such tweets still serve to transmit knowledge. By contrast, scientists' use of uninformative cues, such as partisan emojis in bios, may unnecessarily narrow their audience, intensify affective polarization, and hinder the flow of information in public debate.

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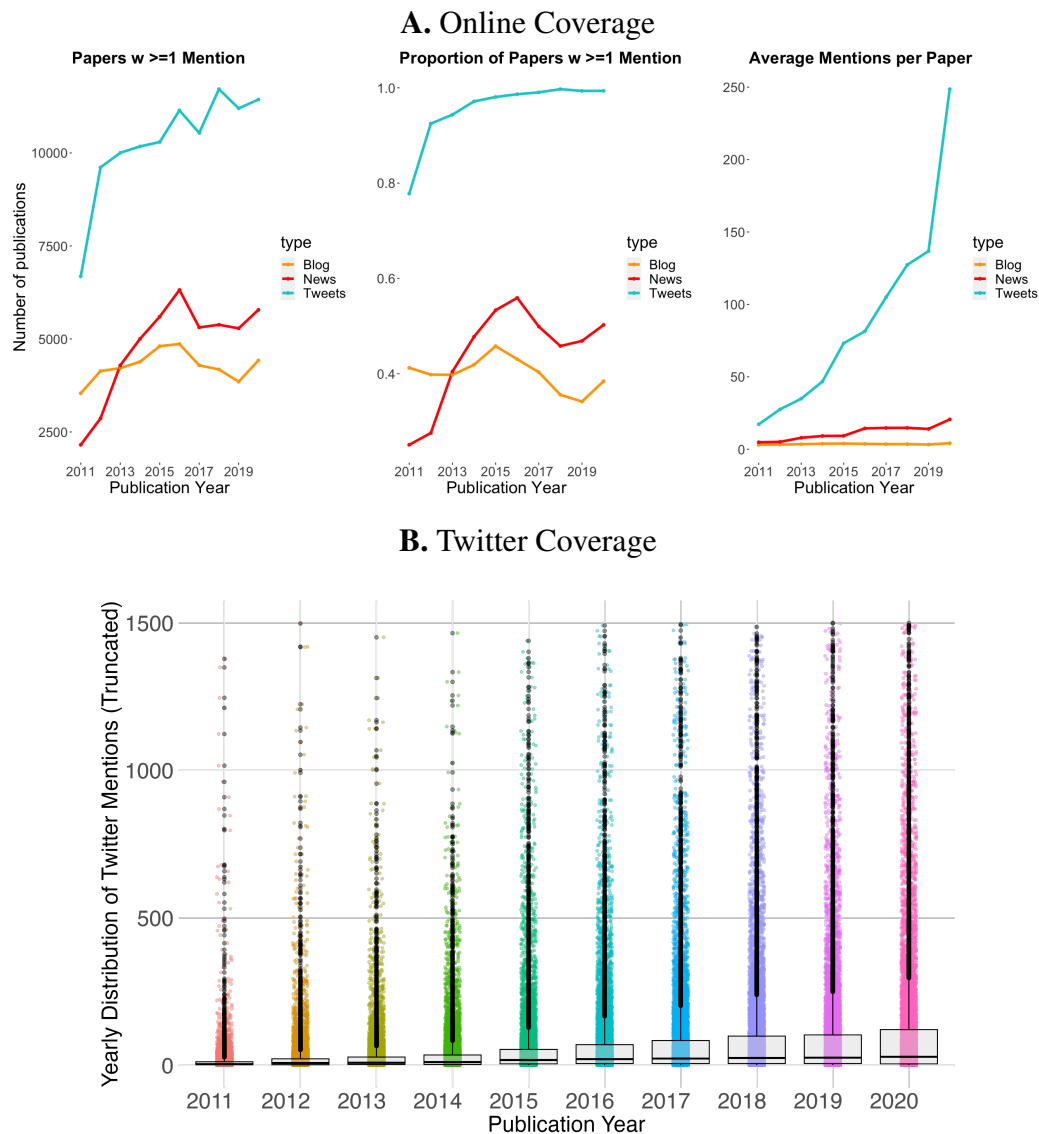
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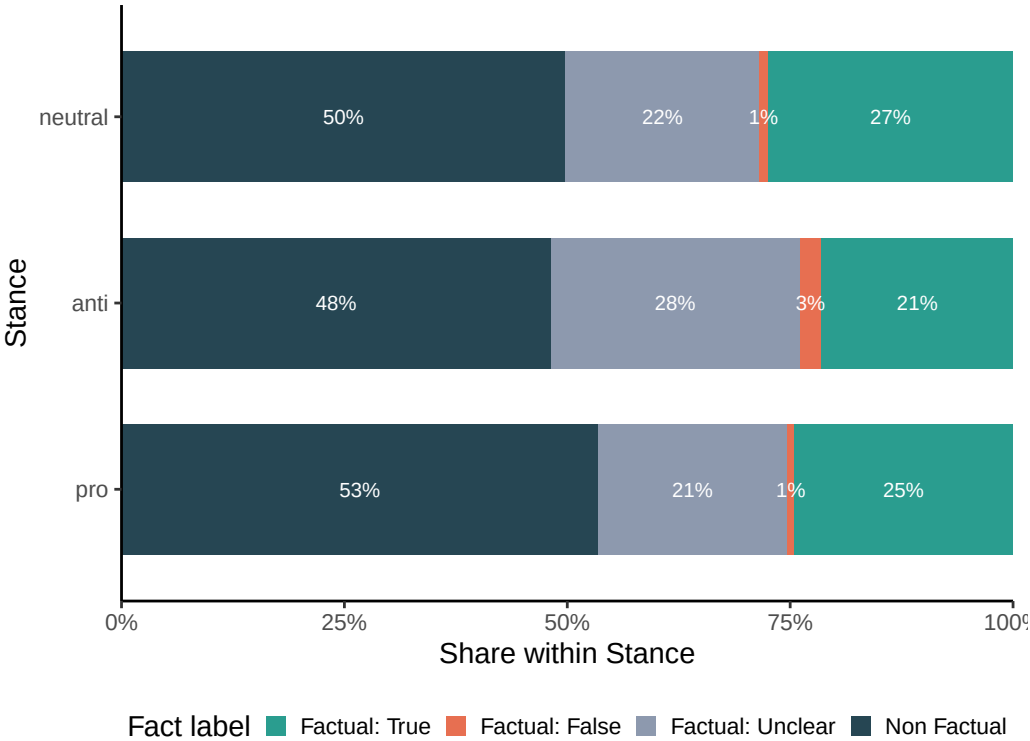
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FIGURE 1.—Online presence of research articles published in general interest journals between 2011 and 2020



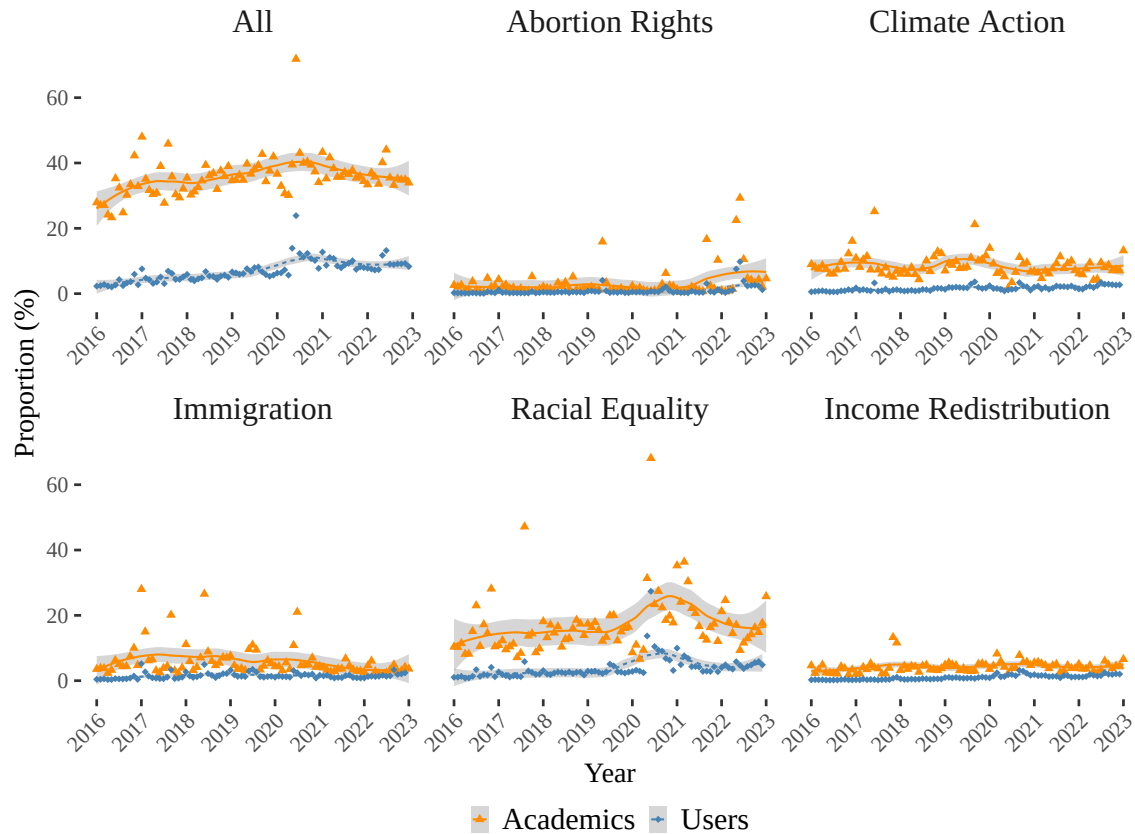
Note: Figure A provides trends in online coverage of scientific articles published in *Science*, *Nature*, *PNAS*, *Cell*, *NEJM*, and *Lancet* between 2011 and 2020. Online appearances across blog posts, newspaper articles, or Twitter are retrieved from Altmetric (accessed on November 10th, 2021). The figure suggests that scientific articles with any online appearances have increased over time, in absolute number (first row), as a proportion of all articles published (second row), and per average number of appearances per published paper (third row). Figure B provides trends in Twitter coverage of scientific articles published in *Science*, *Nature*, *PNAS*, *Cell*, *NEJM*, and *Lancet* between 2011 and 2020. The data present the distribution of Twitter mentions per article, retrieved from Altmetric (accessed on November 10th, 2021). The figure indicates growing online presence, with the distribution of Twitter mentions becoming less skewed towards zero, with a ticker right tale and a rise in high-mention outliers over time.

FIGURE 2.—Distribution of Factual vs Non-Factual Content by Ideological Stance.



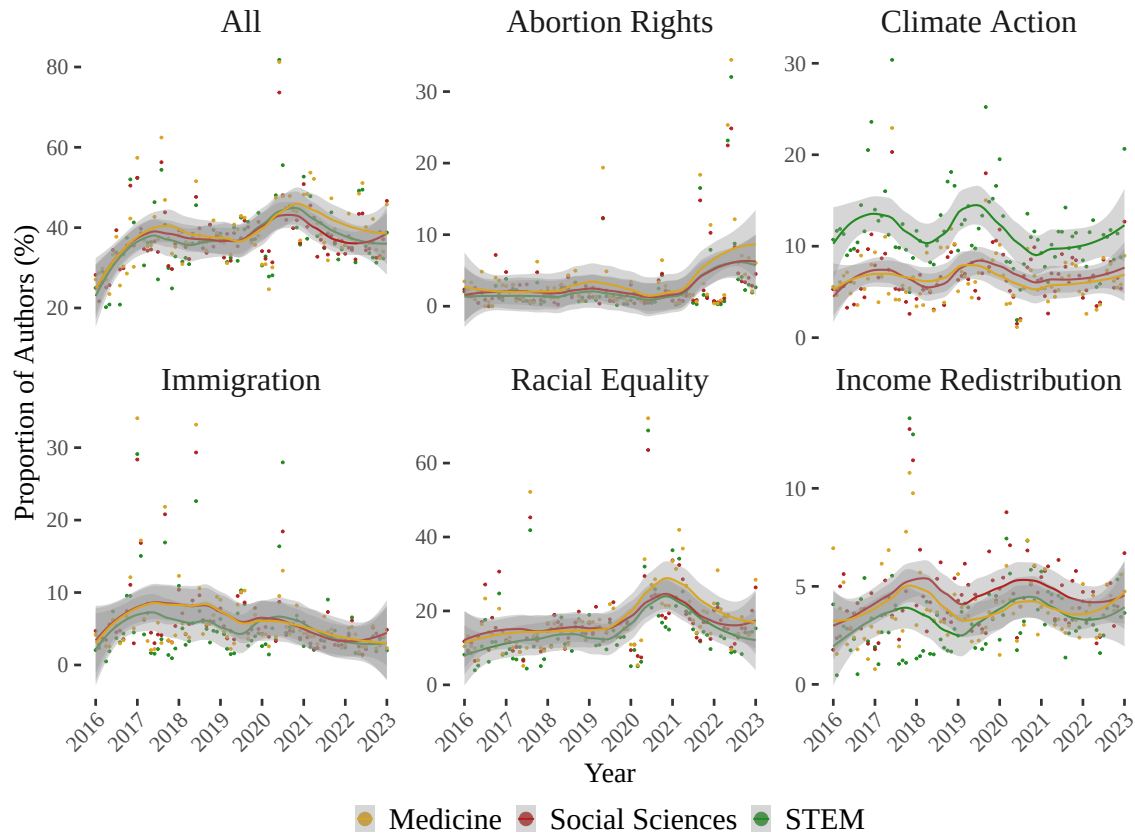
Note: This figure shows the distribution of tweets by academics (2016–2022), highlighting the share of factual (TRUE, FALSE, UNCLEAR) vs non-factual content across ideological stances. Over half of topical tweets (51%) are non-factual, i.e., opinion or commentary which do not contain any verifiable claim. Among factual tweets, 53% are TRUE, 45% UNCLEAR, and 2% FALSE. Conservative-leaning (anti) tweets show relatively more FALSE and UNCLEAR content; progressive-leaning (pro) tweets show a higher share of non-factual content and fewer FALSE classifications. This classification was conducted using GPT-4o-mini, following a structured prompt to label tweets and extract factual claims (full prompt in Appendix Section 4) and helps clarify that a non-neutral stance does not necessarily indicate politicized expression, but may reflect factual reporting aligned with a given issue (e.g., new evidence on climate change). We show this is not systematically the case: while many stanced tweets (particularly in the *pro* category) are grounded in verifiable facts, over half of non-neutral tweets contain no checkable claim at all. Our factual classification thus complements the stance measure by distinguishing between opinion, ambiguous claims, and factual reporting within salient topics.

FIGURE 3.—Proportion of US Users and Academics with a Political Opinion Over Time



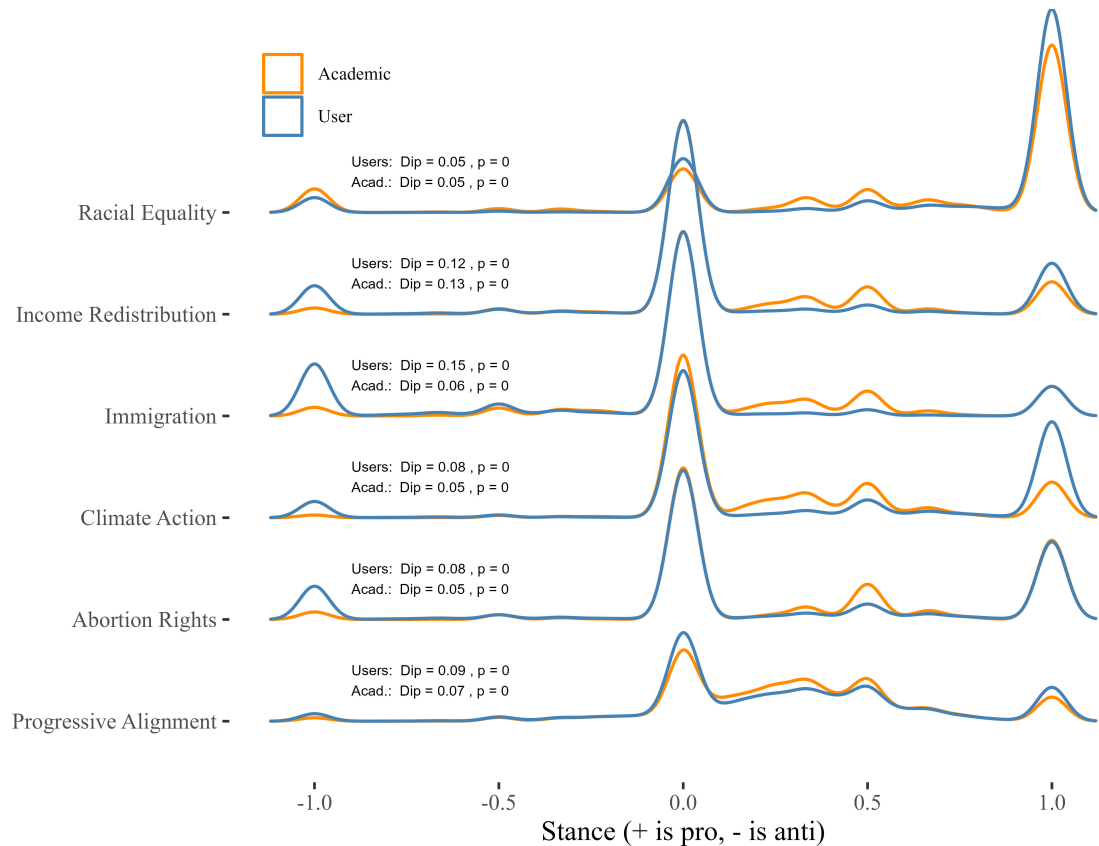
Note: The figure illustrates trends in politicization of conversations by academics and general users on Twitter from 2016 to 2022. Monthly aggregated scatter plots display expressed stance for each topic, with a LOESS applied for trend visualization. Standard errors are depicted in the shaded region. In the "All" panel, around 40% of tracked US academics expressed opinions on predefined political issues, compared to 5-10% of general users. Variations and spikes are observed across topics, with Climate Action and Racial Equality showing the largest disparities. Climate Action witnessed significant declines during the COVID-19 pandemic onset. Mid-2020 saw a surge in attention to Racial Equality, reflecting the outcry after the George Floyd incident. Other topics exhibit stable increasing trends, with occasional short-lived spikes, notably in Abortion Rights around changes in laws in 2022. Immigration discussions, while less frequent, maintained regularity, with heightened attention during the 2016 presidential election.

FIGURE 4.—Proportion of Academics with Political Opinions Over Time by Topic and Discipline



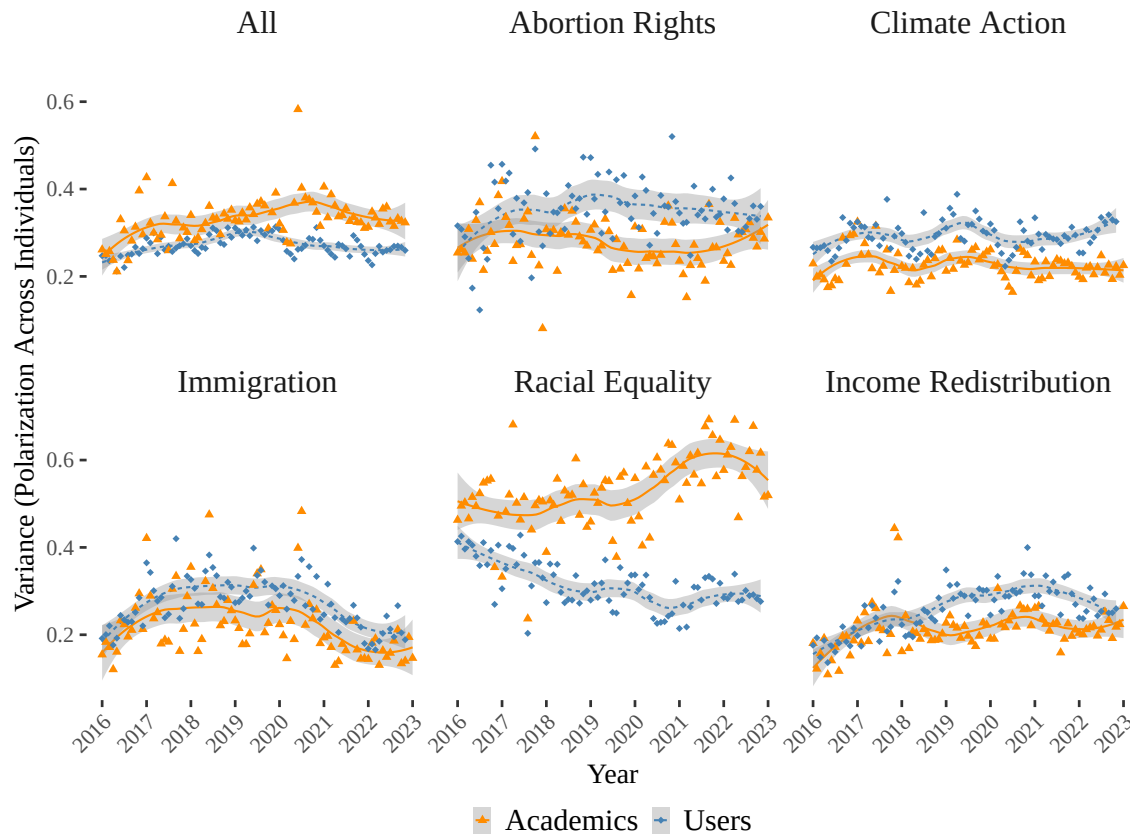
Note: OpenAlex describes "Concepts" as abstract ideas that a work is about: "Concepts are hierarchical, like a tree. There are 19 root-level concepts, and six layers of descendants branching out from them, containing about 65 thousand concepts all told". OpenAlex classifies each work with a high level of accuracy. We combine the 19 root-level concepts into 3 broad categories for ease of comparison: (1) Medicine, (2) Social Sciences, and (3) STEM. We omit Humanities from the time-series depiction given we have only around 100 humanities' academics, which would create unreliable trends. Each work by an author can belong to multiple concepts and a score from 0 to 1 is given to each concept, where values closer to 1 reflect the likelihood the work belongs to that concept. For each academic, we take the average score for all the root-level concepts across all their works from 2016-2022. We then pick the primary concept of that academic as the root level concept with the highest average score. This depicts the proportion of academics within each concept category expressing an opinion about a political topic. Our analysis reveals distinct patterns and spikes across concept categories for two issues: Climate Action and Income Redistribution. STEM academics consistently show significantly higher engagement with Climate Action, almost doubling the frequency of expressions from other fields in any given period. On Income Redistribution, however, STEM academics were notably less vocal than those in Medicine or Social Sciences before 2020, after which the differences narrowed.

FIGURE 5.—Cross-sectional Ideological Polarization across US Users and Academics, 2016-2022



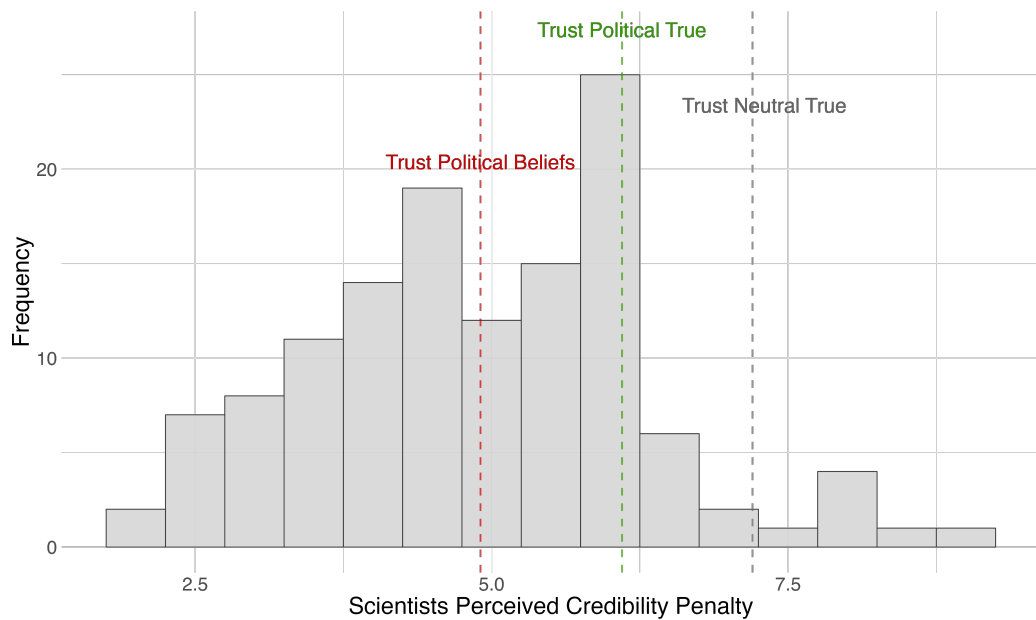
Note: The figure illustrates the density distribution of net stance across various political topics among U.S. academics (in orange) and general users (in blue) from 2016 to 2022. Topics include Income Redistribution, Climate Action, Immigration, Abortion Rights, and Racial Equality, with "Progressive Alignment" representing the average stance across these topics. The x-axis represents the net stance, where positive values indicate a pro-stance and negative values indicate an anti-stance. The y-axis indicates different topics, with density distributions shown as ridgelines. Each ridgeline highlights where individuals tend to cluster in their expressed opinions. Black vertical lines within each distribution represent the mean net stance for each topic. Hartigan's Dip Test identifies multimodal distributions, suggesting distinct ideological camps. The dip statistic and corresponding p-value are annotated for each topic, demonstrating statistically significant multimodalities for both groups. Compared to academics, general users tend to cluster more around extreme viewpoints, especially towards the conservative (-1) side on most issues. Academics, on the other hand, have a larger mass in the moderate liberal/progressive range. An exception to this is the issue of race, where general users show more consensus, with a larger mass towards pro-racial equality. Both groups exhibit statistically significant multimodal distributions (p-values of 0), with users showing slightly higher dip statistics on average, indicating more pronounced ideological polarization.

FIGURE 6.—Issue Disagreement Across Academics By Topics Over Time



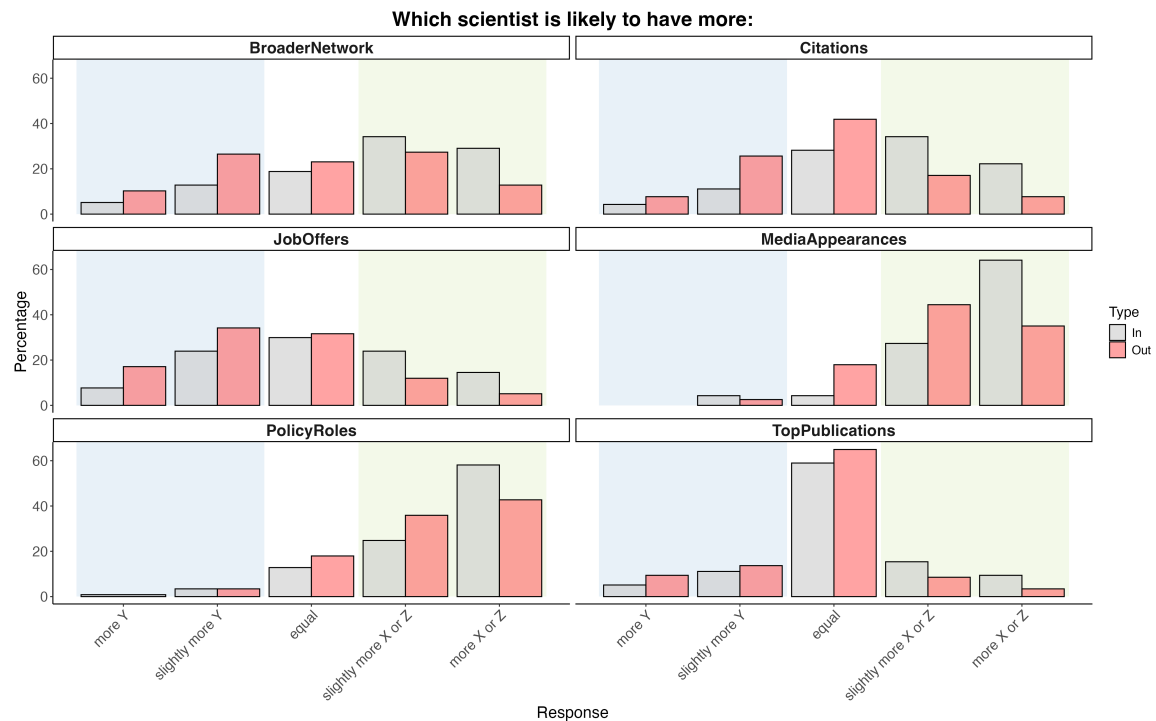
Note: The figure measures ideological polarization by computing the variance of net stance held by a population on predefined political topics. Variance across users reflects the dispersion of political opinion on each topic, providing a continuous measure of ideological disagreement. Aggregating across topics yields an overall measure of political variance. Similar to the increasing trend in expression of opinion, polarization increases between 2016 and 2022, raising concerns about its impact on scientific consensus-building and public trust in scientific expertise. Race exhibits wider debate among academics relative to the general population. Disparities in ideological disagreement are observed across topics, with gaps widening, particularly around the pandemic period for Immigration, Income Redistribution, and Abortion, but also closing by the end of 2022. The gap persists for Climate Action throughout the sample period and grows at the end of 2022.

FIGURE 7.—Scientists' Beliefs about Credibility Penalty



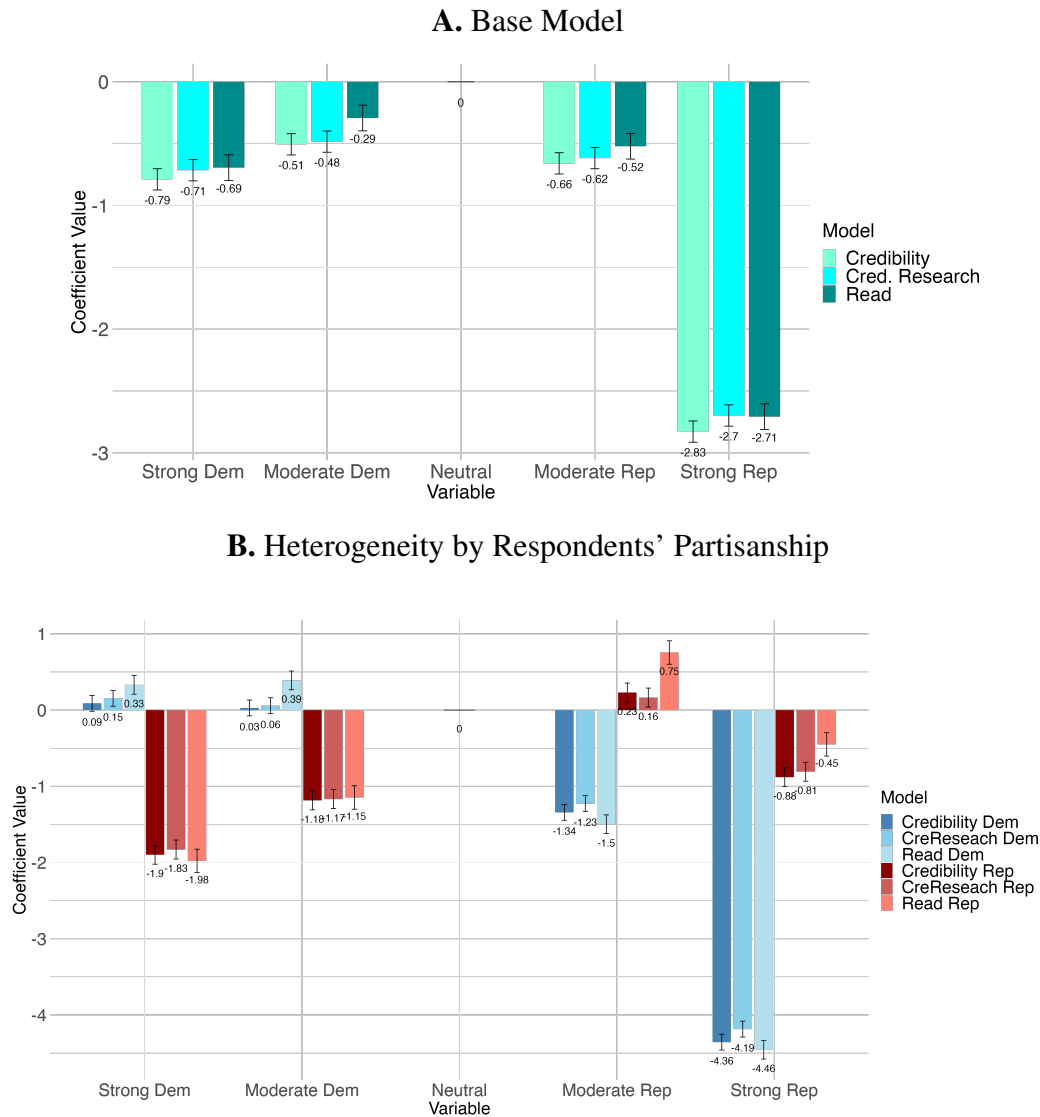
Note: This figure reports the distribution of responses from 128 scientists recruited on Prolific to the following question: "What do you think is the reported level of trust for scientists who do express political opinions on social media?". Prior to asking the question, we informed the academic respondents that we had surveyed a representative sample of the U.S. on their perceived credibility of scientists, distinguishing between those who had expressed political opinions online and those who had not. We anchored our scientists' beliefs on the public perceived credibility for scientists who *do not* express political opinions online (indicated by the grey dashed line). The average answer of our sample of scientists is shown by the red dashed line and is significantly lower than the true value obtained in the main survey of a representative sample of U.S. respondents, which is indicated by the green dashed line.

FIGURE 8.—Scientists' Perceived Costs and Benefits from Public Political Expression



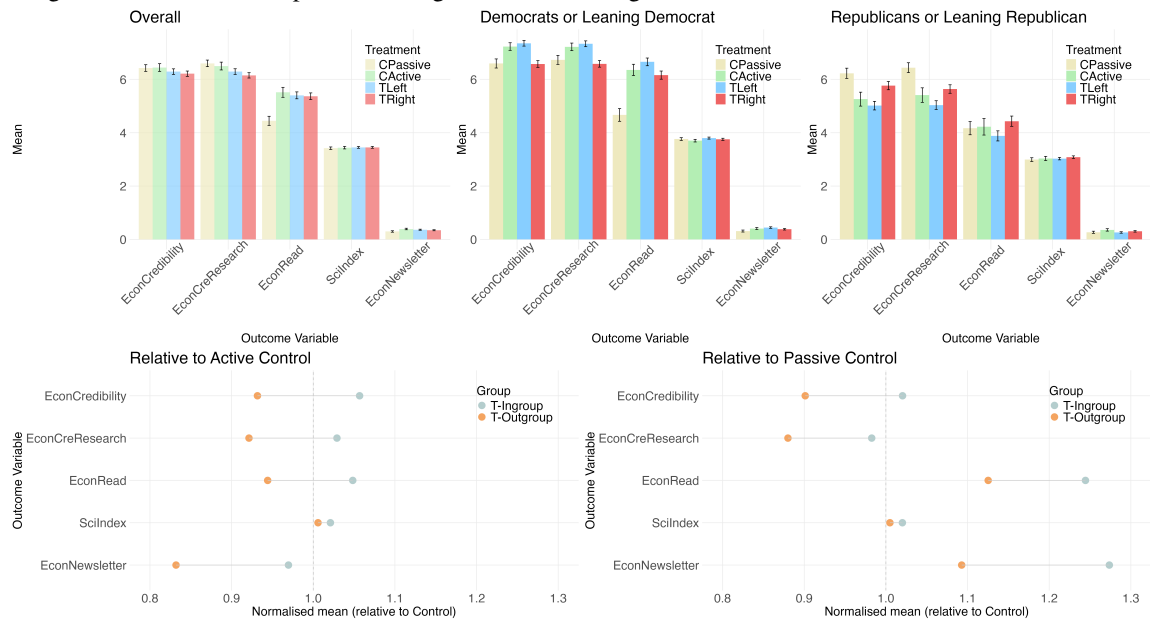
Note: The figure illustrates scientists' perceptions of professional benefits based on public political expression. Respondents compared the outcomes for Scientist X (grey), who shares opinions on politically charged topics related to their research, and Scientist Z (red), who shares opinions on politically charged topics unrelated to their research, against Scientist Y, who refrains from such expression (respectively, either related or unrelated to their research). The evaluated benefits include network size, citations, academic job offers, media appearances, policy roles, and top publications. A distribution skewed towards X or Z (green area) reflects perceived benefits from political expression, while a denser left tail towards Y (blue area) indicates perceived costs. Results suggest clear benefits for media appearances and policy roles, milder benefits for network size and citations, and ambiguous or slight perceived costs for top publications and academic job offers.

FIGURE 9.—Impact of Scientists' Political Expression on Perceptions from the General Public. Credibility and public Willingness to Read Peak at Neutral, with a Monotonic Penalty for Scientists Political Expression to the 'Left' and 'Right' of Neutral.



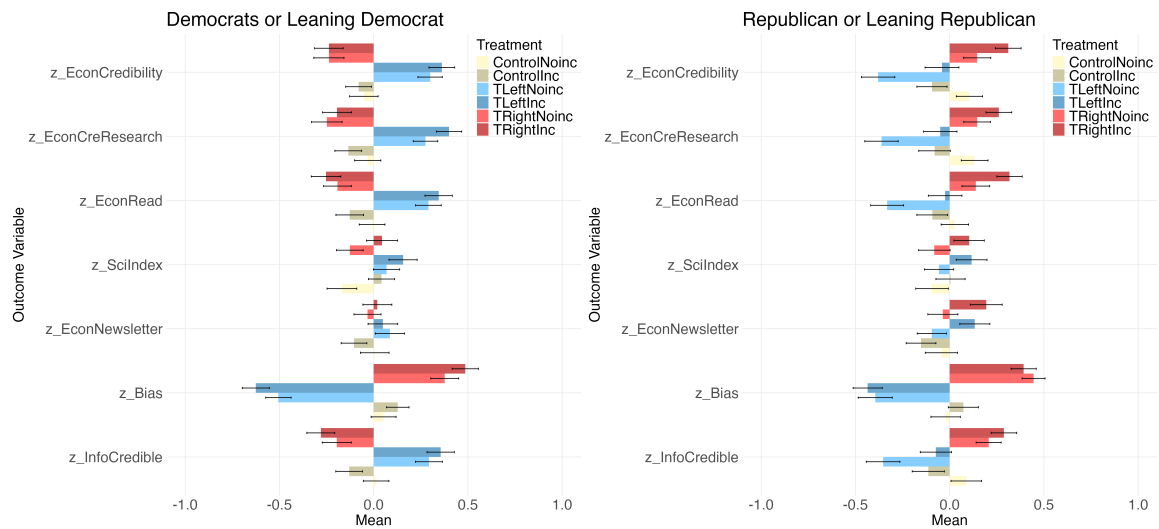
Note: Coefficients are obtained by regressing scientists' characteristics on respondents' perceived credibility or willingness to read content from scientists. The x-axis represents different political affiliations of scientists, estimated by indicator variables for "Strong Republican", "Moderate Republican", "Strong Democrat", or "Moderate Democrat", with "Neutral" as the excluded category. The y-axis shows the coefficient values indicating the impact on credibility and willingness to read. The data reveals a peak in credibility for neutral scientists, with a decline for both left- and right-leaning scientists. Standard errors are clustered at the individual level. Additional regressors include indicator variables to control for other scientist characteristics: institutional affiliation (Harvard, UC Berkeley, or Chicago, versus Arkansas or Connecticut), field of research (Medicine, Mathematics, Engineering, Economics, or Literature), seniority of role (Full professor or Assistant Professor), and gender (male or female). (N = 1704, 940 Dem. or Lean Dem., 745 Rep. or Lean Rep., 19 Other leaning.)

FIGURE 10.—Separating the Effect of Communicating Salient Research from a Scientists' Pure Political Signal on Respondents' Perceived Credibility. Any Signal Impacts Scientists' Credibility. Effects are Moderated by the Congruence between Participants' Leaning and Scientists' Signal.



Note: Figures show results of our second experimental task where respondents are divided into four groups: in the *passive control* group, respondents are exposed to an economist who neither advertises own research in a politically salient issue nor signals any political affiliation; respondents in the *active control* group are exposed to the profile of an economist advertising own research in a politically salient issue with no political signal; respondents in the *treatment left (right)* group are exposed to an economist advertising their research in a politically salient issue together with a left (right) political signal. The politically salient research is favourable to a democrat leaning narrative. After viewing one of the four profiles, we collect the following outcome variables for each respondent: their perceived credibility of the economist, their perceived credibility of the economist's research, their willingness to read an opinion from the economist, their intention to sign up for a newsletter containing opinions from a similar profile, and a composite index of general trust in science. *Democrats* show higher credibility when exposed to politically aligned research. The left political signal increases perceived credibility, while the right signal reduces it. Similarly, willingness to read is higher with politically aligned research; the left signal increases it, while the right signal decreases it. *Republicans* exhibit significantly reduced perceived credibility when exposed to misaligned politically salient research, especially with a left signal, though less so with a right signal. Willingness to read is highest with a congruent right signal and lowest with a left signal. For both sub-samples, newsletter sign-up and overall trust in science move similarly, but changes are less pronounced. Additionally, normalising our group averages relative to the active or passive control, at the bottom, we observe that in-group respondents (when scientist signal and respondents leaning align) perceive significantly higher credibility of scientists and their research, are more willing to read their opinion, and are more likely to sign up to the newsletter, relative to out-group respondents. (N = 1704, 940 Dem. or Lean Dem., 745 Rep. or Lean Rep., 19 Other leaning.)

FIGURE 11.—Isolating the Effect of Strategic Considerations behind Scientists' Political Expression on Respondents' Perceived Credibility: Making *Visibility Incentives* Explicit. Subjects Respond to Scientists' Incentives, but Updating Depends on Partisan Congruence.



Note: Figure shows results of the experimental task isolating the effect of explicit strategic motives behind scientists' political expression. Respondents, selected among Twitter users, were randomly assigned to a 2×3 design: Varying by *Tweet leaning* (Neutral, Left, Right) and *Promotion information* (No Promotion vs Promotion). In the Promotion condition, respondents were told that *we (the researchers)* would promote an economist's tweet and that the economist knew this *before* posting; no such information was provided in the No-Promotion condition. Outcomes include credibility of the economist, credibility of the economist's research, willingness to read related content, a Trust in Science Index, perceived ideological bias of the economist, and credibility of the information in the tweet. Bars plot conditional means of standardized outcomes; whiskers show 95% confidence intervals. Promotion information does not produce a uniform penalty. Among Democrats, it amplifies partisan reactions. *Relative to No-Promotion*, perceived credibility is higher for *promoted left-leaning* tweets and lower for *promoted right-leaning* tweets, while neutral promotion has no impact. Among Republicans, promotion raises perceived credibility for *promoted right-leaning* tweets and attenuates the penalty for *promoted left-leaning* tweets; promotion of a neutral tweet is penalized. General trust in science increases for Democrats when facing left (and neutral) promoted content, and for Republicans when facing either left- or right-leaning promoted content. Perceived ideological bias of the economist shifts with tweet slant as intended, confirming the manipulation. (N = 2,000; 1,081 Dem./Lean Dem.; 906 Rep./Lean Rep.; 13 Other.)